

Digital Twin: Bringing MEP Models to Life

David A. Fink Digital Manager - Henning Larsen Architects

Alejandro Mata Automation Manager - Rambøll





About the speaker

David A. Fink

Digital Manager – Henning Larsen Architects, Denmark

- M. Arch – University of Michigan
- Registered Architect – Massachusetts
- Founding member of BIMCPH and DUGCPH
- Loves playing with all things digital, running and mountain biking

Global and Local Expertise

60 Years
Founded in 1959

6
Global Offices

25
Countries

Scandinavian
Copenhagen Based
International Reach

300+
People,
36 nationalities

**Henning
Larsen** —



Our Services



Architecture



Interior and product design



Graphic design and wayfinding



Sustainability



Landscape and urban design



About the speaker

Alejandro Mata

Automation Manager – Rambøll Denmark

Integrated Digital Solutions

MSc. Civil Engineering | HVAC

Digitalization and BIM development for HVAC projects



RAMBOLL IN BRIEF

- Independent engineering and design consultancy and provider of management consultancy
- Founded 1945 in Denmark
- 14,000 experts
- Close to 300 offices in 35 countries
- Particularly strong presence in the Nordics, the UK, North America, Continental Europe, Middle East and Asia Pacific
- EUR 1.4 billion revenue
- Owned by Rambøll Fonden

RAMBOLL

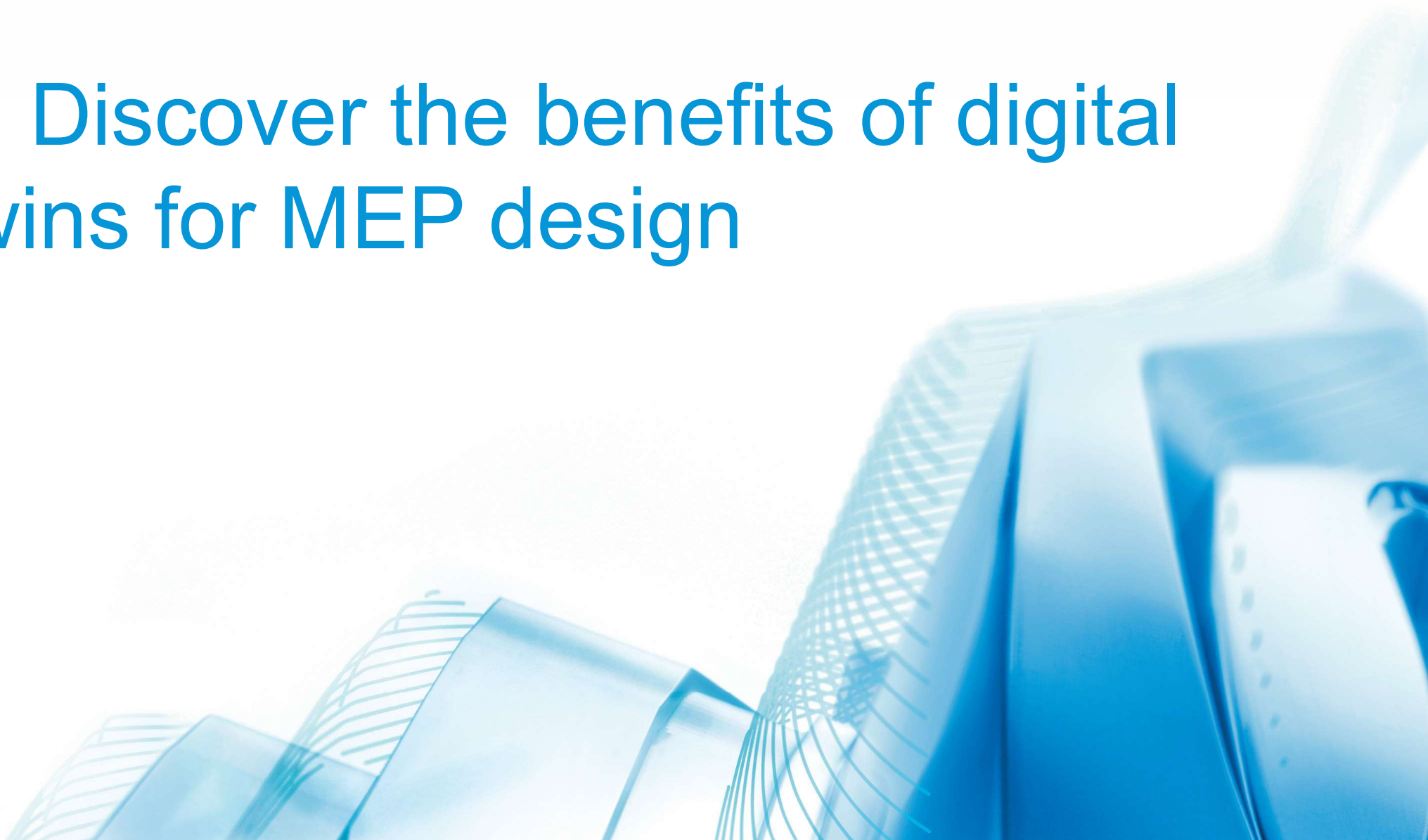
INTEGRATED DIGITAL SOLUTIONS



Learning Objectives

1. Discover the benefits of digital twins for MEP design
2. Learn about bringing together MEP models and IoT sensors
3. Learn how to visualize live IoT data with BI dashboards
4. Discover the different types of IoT devices available for MEP systems

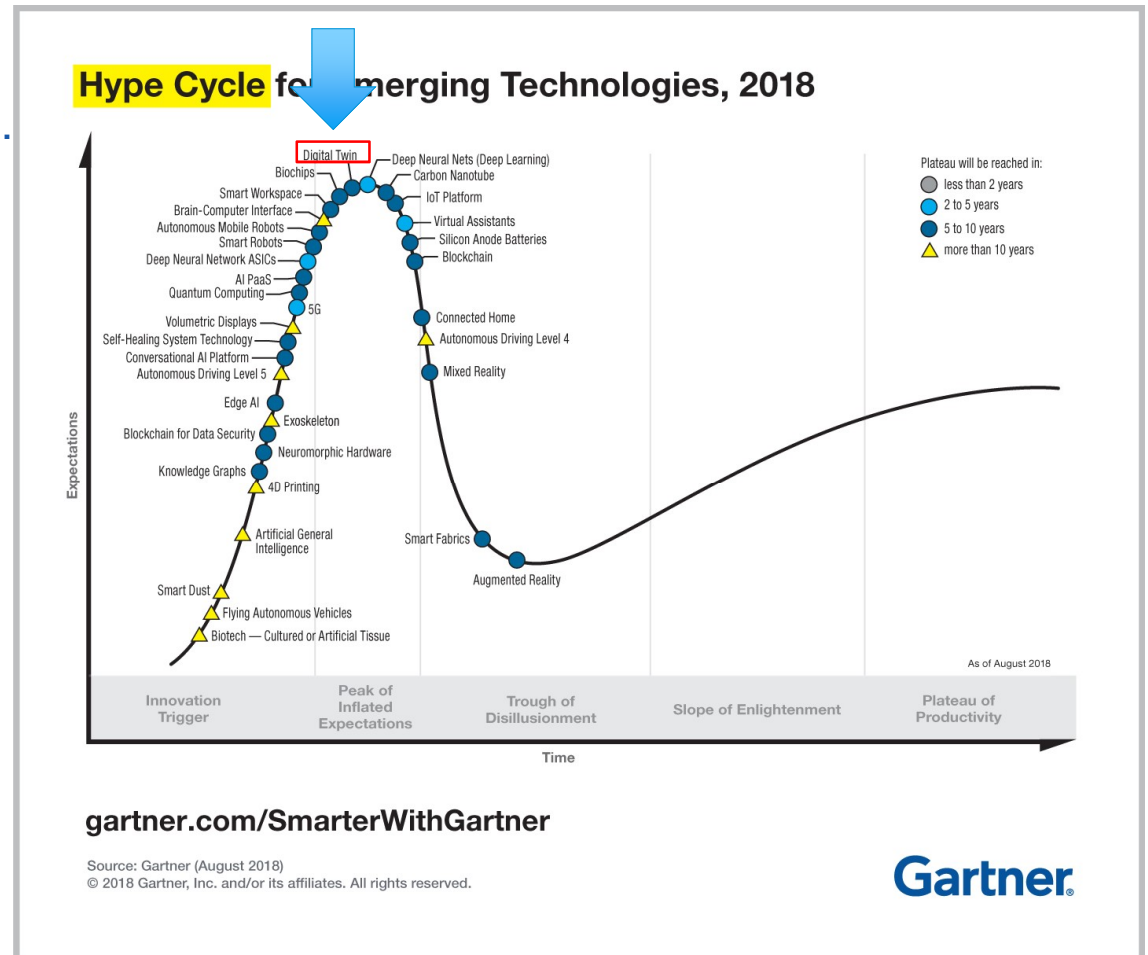
1. Discover the benefits of digital twins for MEP design



Why Digital Twin?

Other Industries are already doing it...

- Manufacturing
- Space
- Aerospace
- Urban Planning
- Automotive
- Healthcare
- Asset Management
- Financial



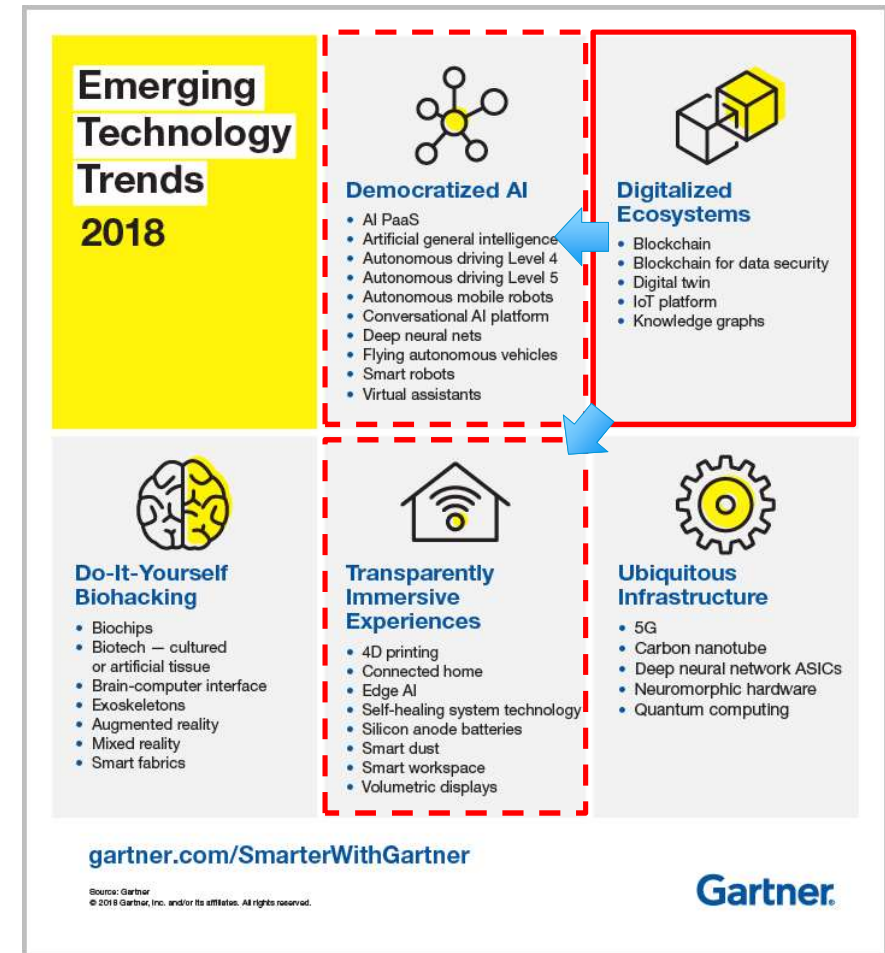
Why Digital Twin?

Digital Twin part of Digitalized Ecosystems

- Democratized AI
- Transparently Immersive Experiences



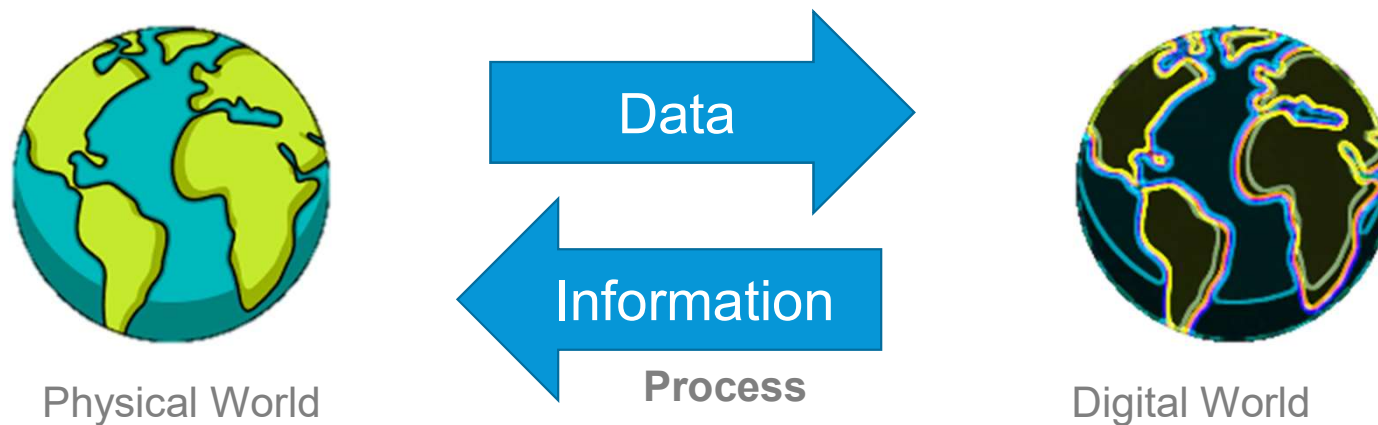
Thomas Gregersen AU 2017



What is Digital Twin?

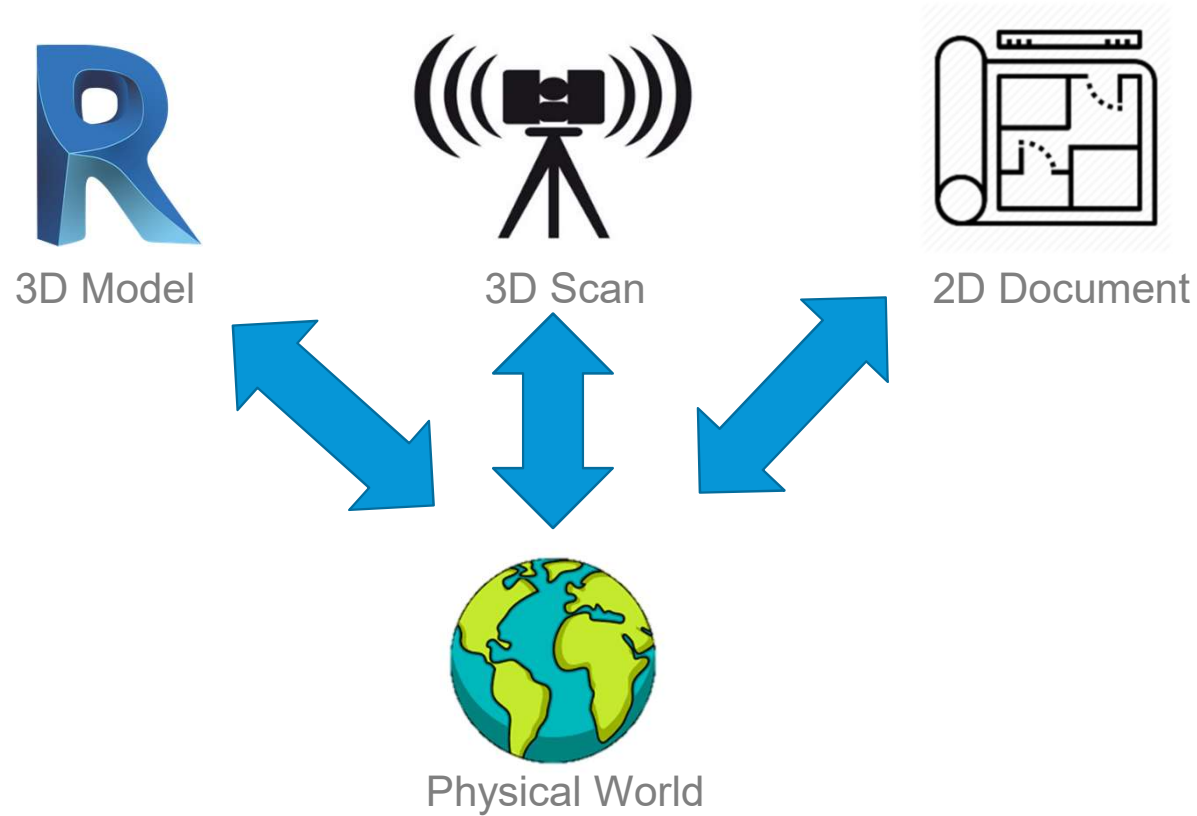
Digital Twin is a set of virtual information constructs that fully describes a potential or actual physical manufactured product from the micro atomic level to the macro geometrical level.

-Michael Grievers



Linking the Physical with the Digital

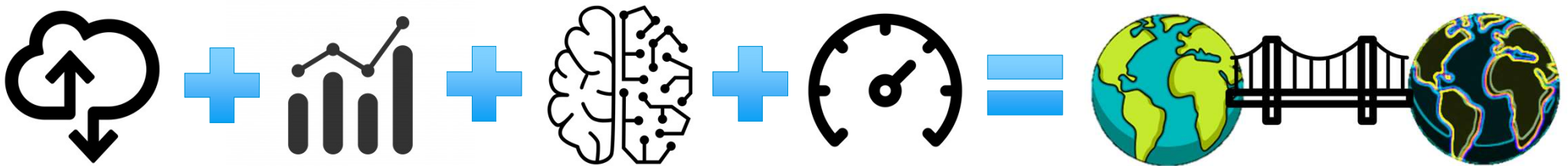
Making the connection between two environments



Merging Multiple Technologies

Digital twins combine a number of techniques:

- Streaming sensor data
- Analytics
- Machine learning or Artificial Intelligence
- Enabling knowledge graph representations of the world
- Creating interfaces that can bridge the digital and the physical worlds.



Mirrored Systems

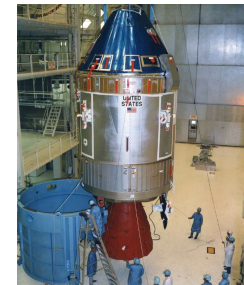
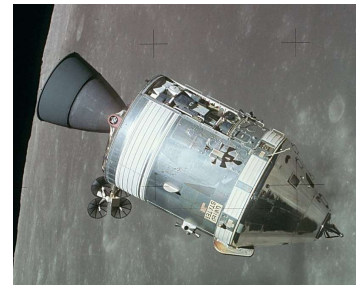
NASA

"Analog Twin"

Modifications were made to an identical physical representation of the actual system in use.

Spacecraft = System

- Receiving sensor data
- Analyzing the data
- Dash boarding data
- Responding to the data



Digital Twins in use

Formula 1

Real time monitoring the performance of the system on the track.

Modifications made to a digital copy of the system in use.

Car = System

Receiving sensor data

Analyzing the data

Dash boarding data

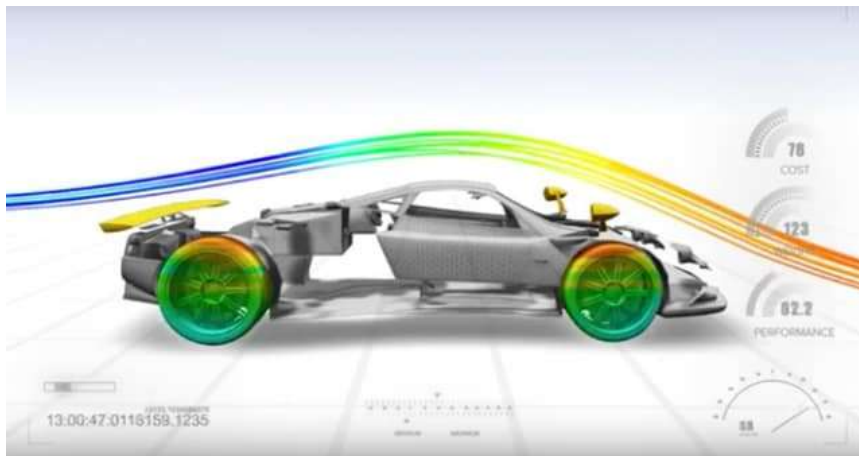
Responding to the data



Types of Digital Twins

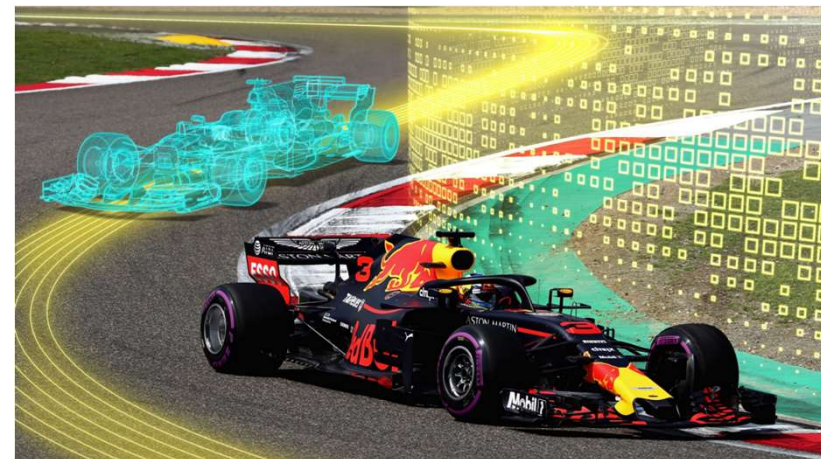
DIGITAL TWIN PROTOTYPE

- Prototype of a physical system
- Contains all the required information for creating the system

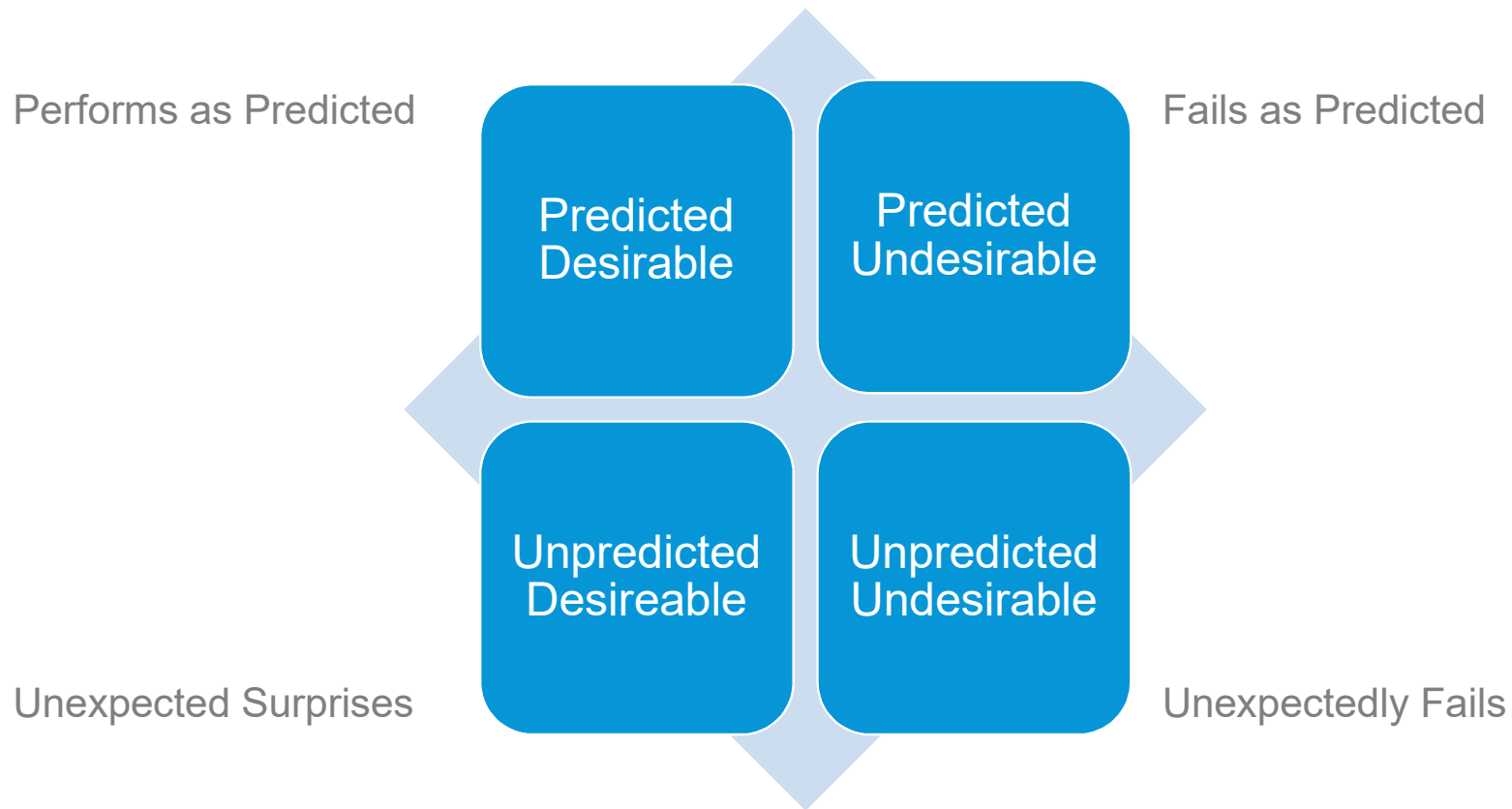


DIGITAL TWIN INSTANCE

- Describes a specific corresponding physical system.
- The prototypical digital twin is linked to the physical system throughout the lifecycle of the system.



Evaluating Digital Twin



Evaluating Digital Twin – MEP Example

Performs as Predicted



Predicted
Desireable

Predicted
Undesirable

Fails as Predicted



Unexpected Surprises

Unpredicted
Desireable

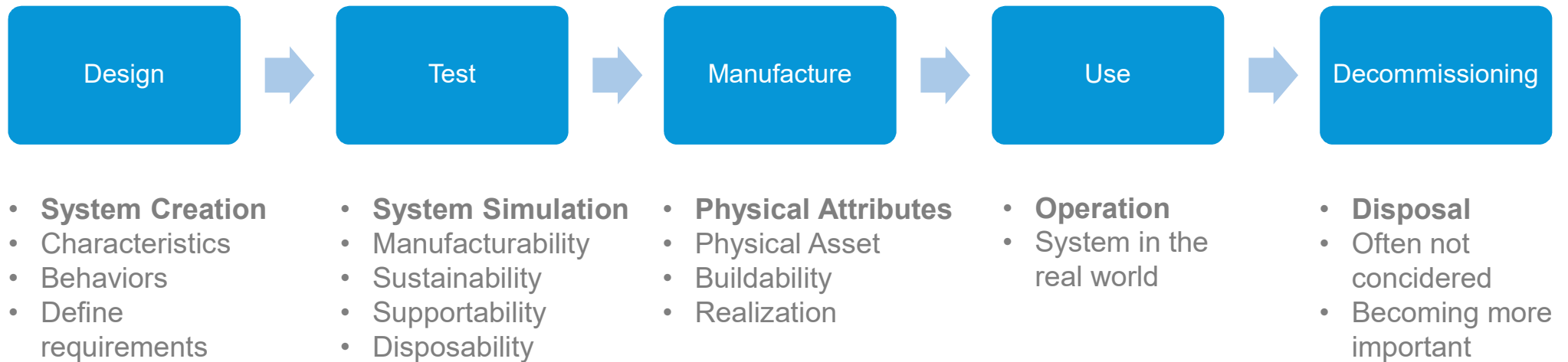
Unpredicted
Undesireable



Unexpectedly Fails

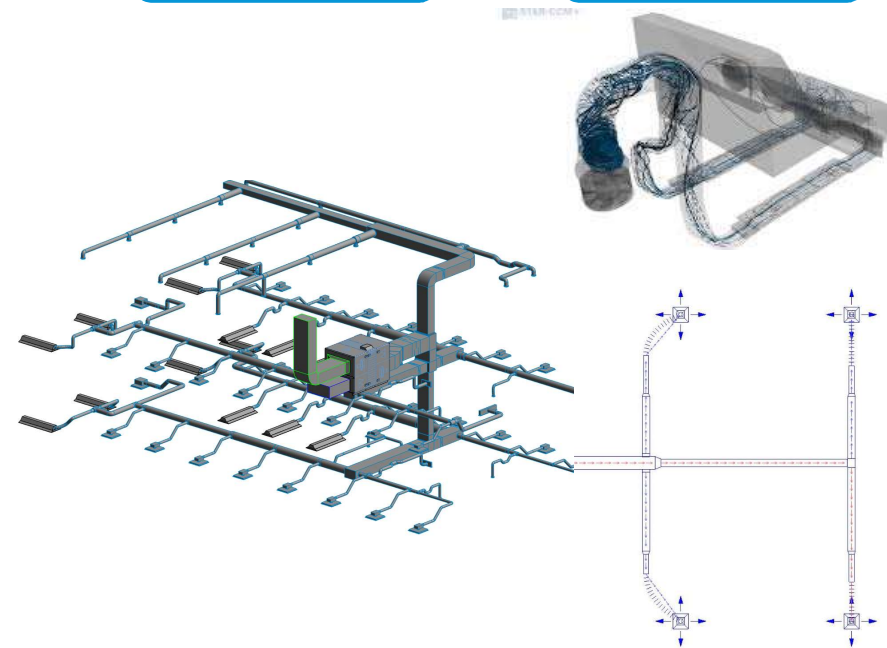
Digital Twin – Process

Lifecycle of a system from design through decommissioning



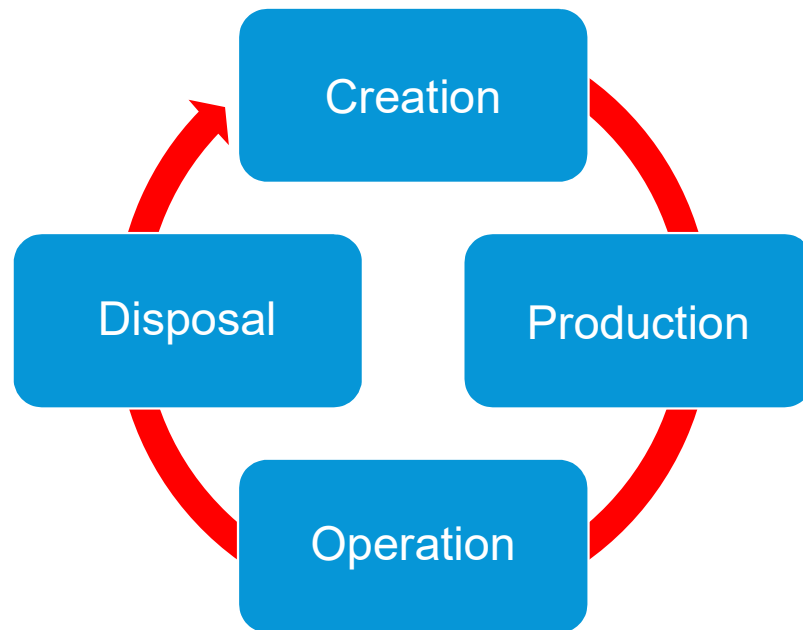
Digital Twin – MEP Process Example

Lifecycle from design through decommissioning

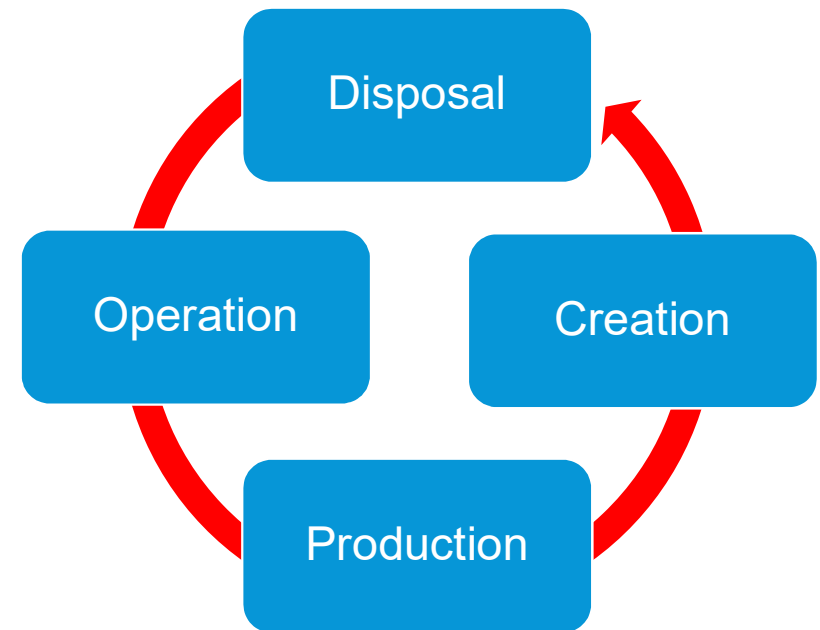


Process

Digital Twin Data flows in Reverse

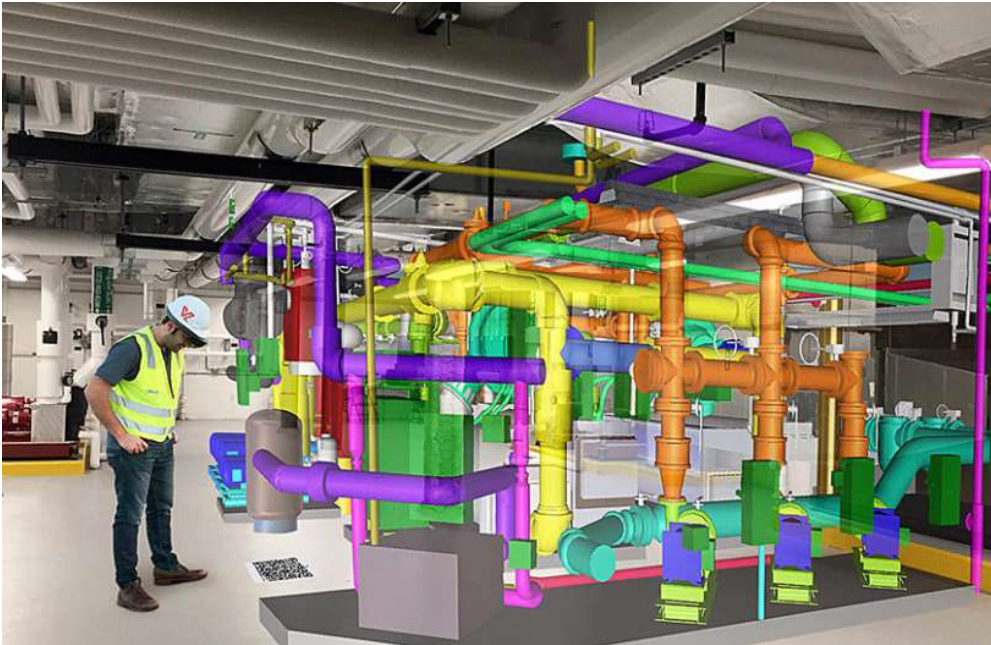


Physical Creation – Design Flow

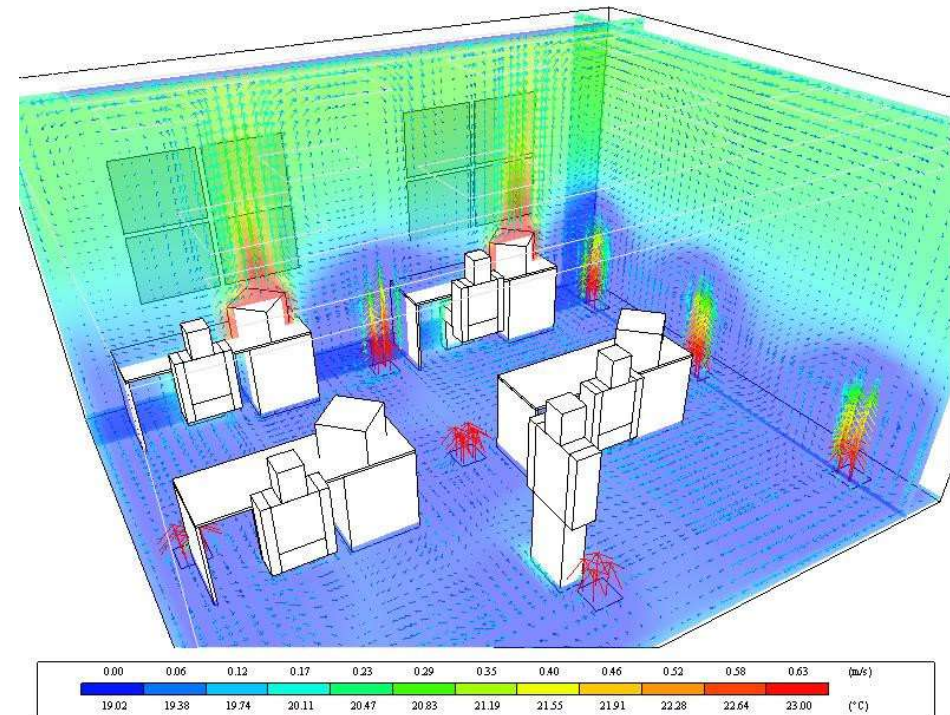


Digital Twin – Design Flow

Process - Example



Physical Creation – Design Flow



Digital Twin – Design Flow

Benefits of Digital Twin

Higher occupancy rate

Up to **3.5%** higher occupancy by using BIM



Source: Handbook for the Introduction of Building Information Modeling by the European Public Sector

Improve budget reliability

Up to **40%** decrease in non-budgeted change-orders



Source: CIFE, Center for Integrated Facility Engineering, Stanford University

Financial opportunity for digitalizing engineering, construction and operations processes to be in the range of

10 – 20%

of capital project expenditure



Source: Handbook for the Introduction of Building Information Modeling by the European Public Sector

Faster project delivery

Up to **7%** shortened project timeline



Source: CIFE, Center for Integrated Facility Engineering, Stanford University

Basis for life cycle cost optimization

Up to **9%** lower operational costs



Source: CIFE, Center for Integrated Facility Engineering, Stanford University

Higher building quality

Up to **3.5%** higher occupancy rate



Source: CIFE, Center for Integrated Facility Engineering, Stanford University

Benefits of Digital Twin



Benefits for Society – Mapping the World solar energy potential



How Project Sunroof Works

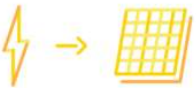
Your own personalized solar savings estimator, powered by Google Earth imagery.



1

Search for your home

We use Google Earth imagery to analyze your roof shape and local weather patterns to create a personalized solar plan.



2

Personalize your solar analysis

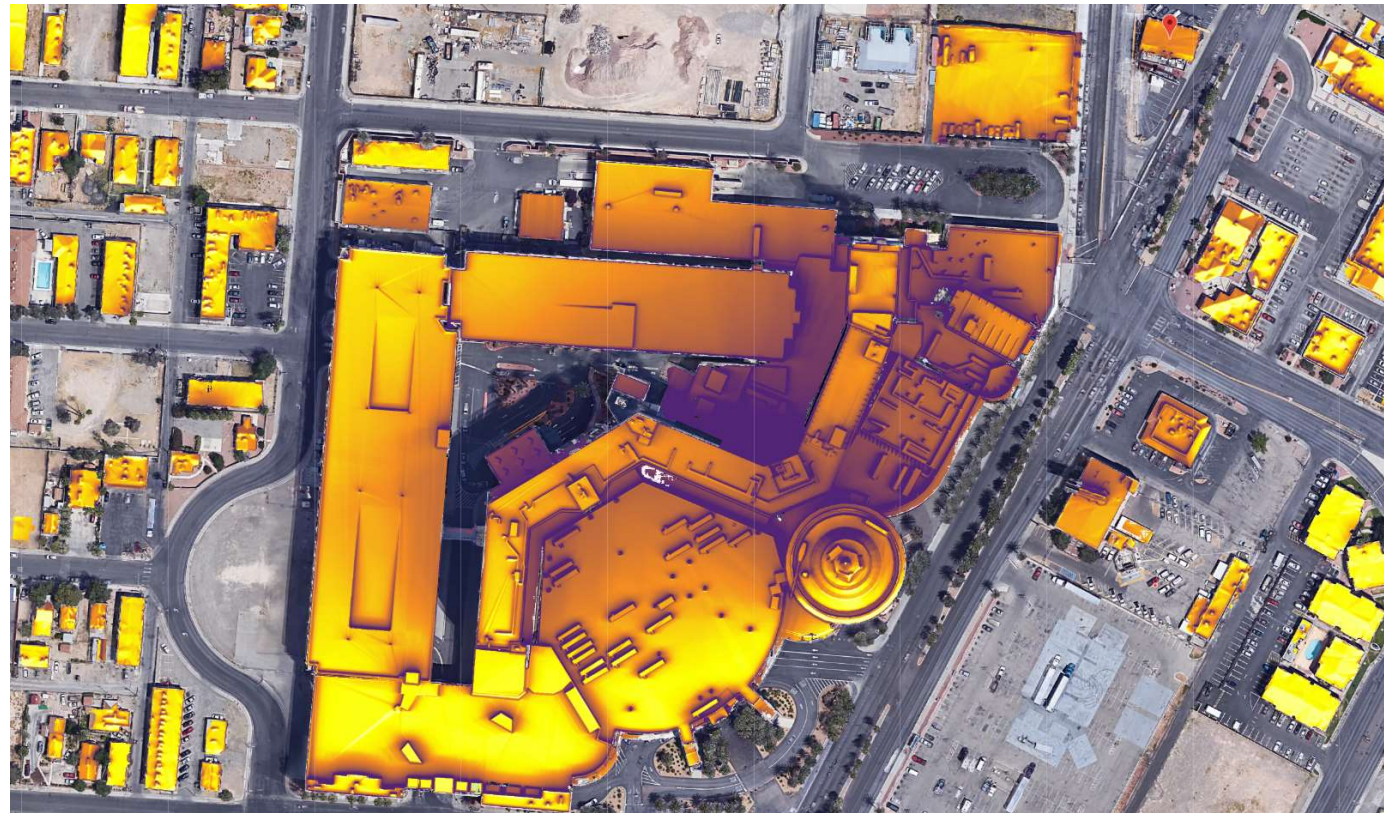
Adjust your electric bill to fine-tune your savings estimate and the recommended number of solar panels for your home.



3

Compare finance options

Compare loan, lease, and purchase options for your solar panels based on your results.



Benefits for Enterprises – Smart Buildings

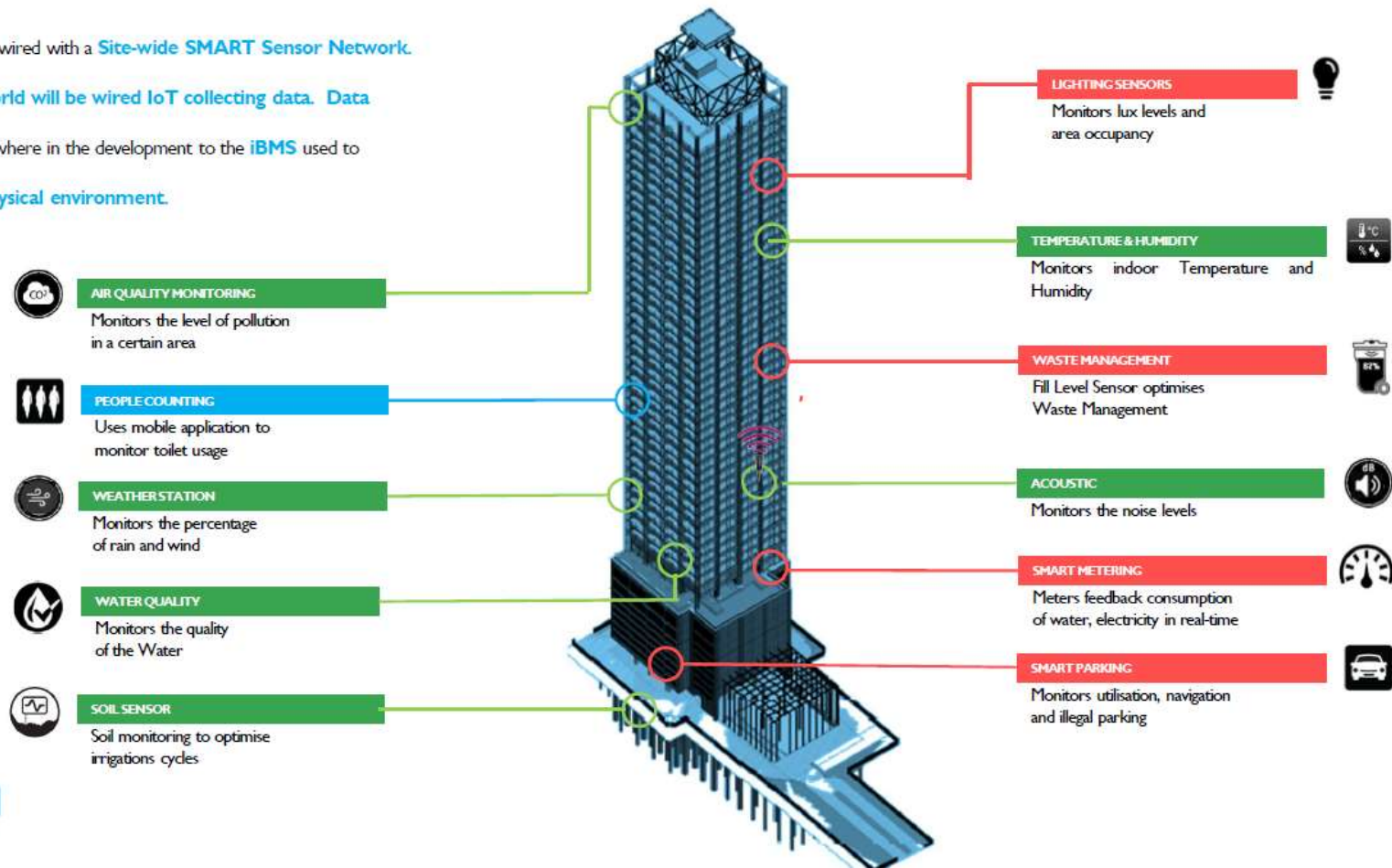
SENSOR NETWORK

Buildings will be wired with a **Site-wide SMART Sensor Network**.

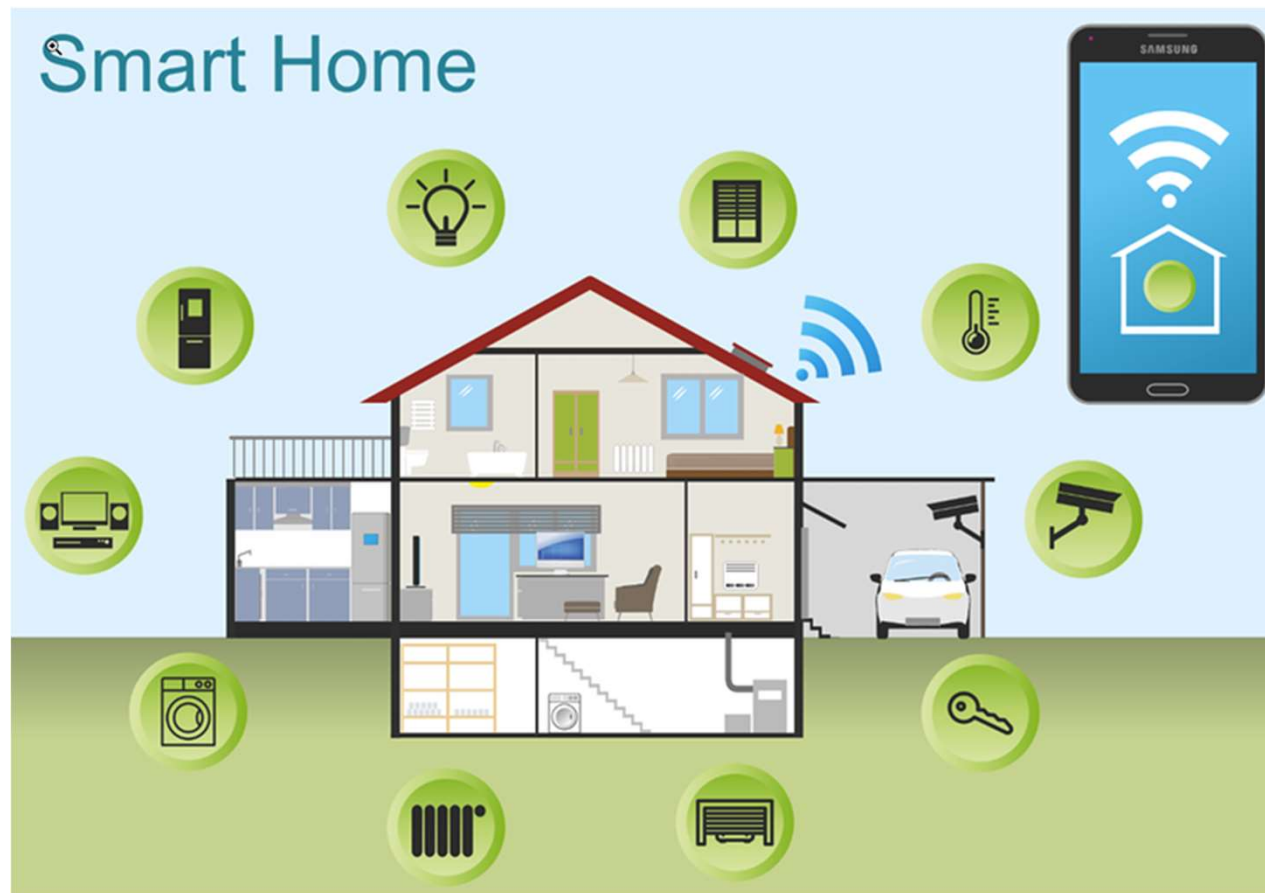
The physical world will be wired IoT collecting data. Data

flows from everywhere in the development to the **iBMS** used to

optimise the physical environment.



Benefits for Private Owners – Smart Homes



2. Learn about bringing together MEP models and IoT sensors

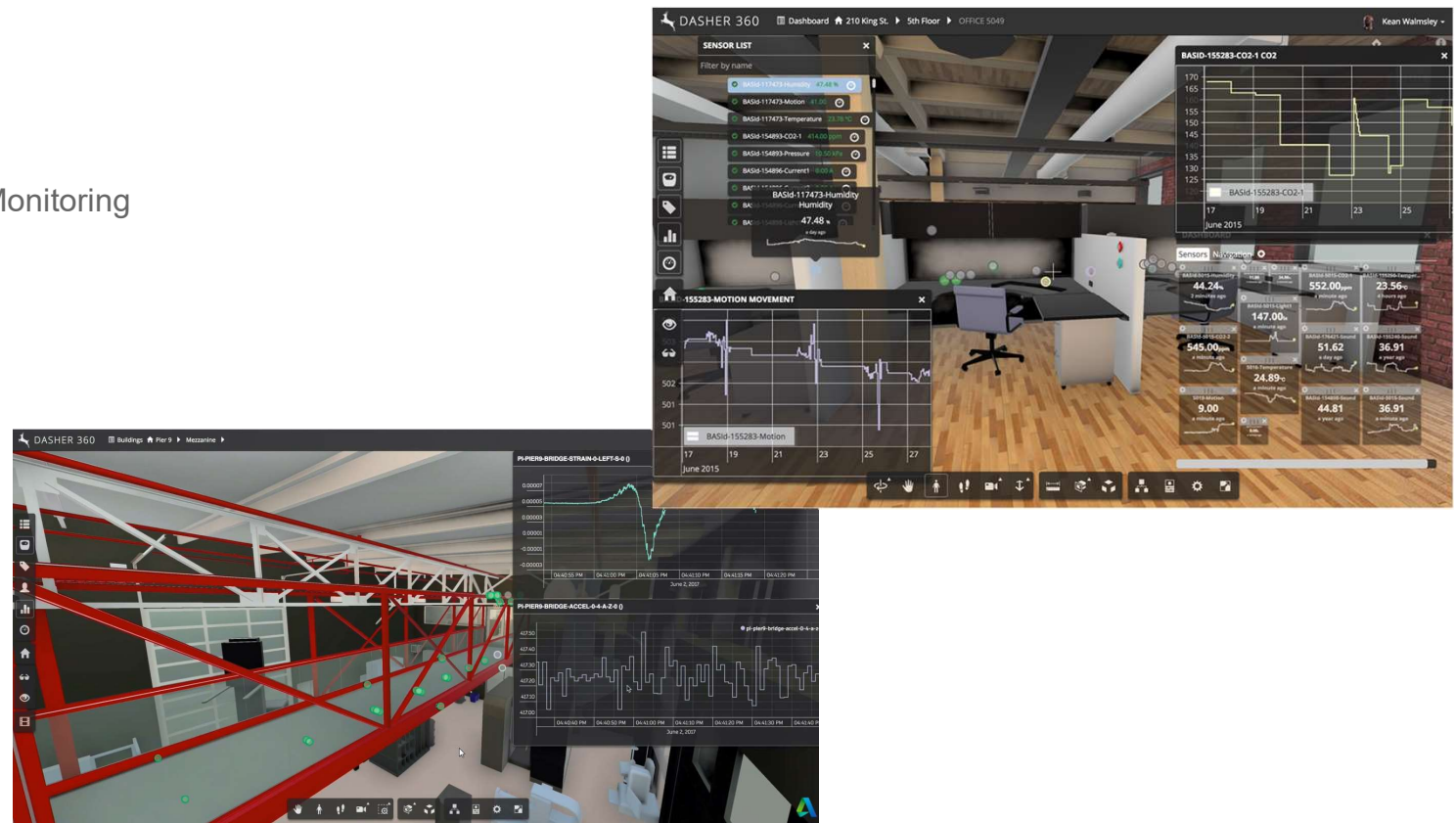


Autodesk Dasher 360 Projects

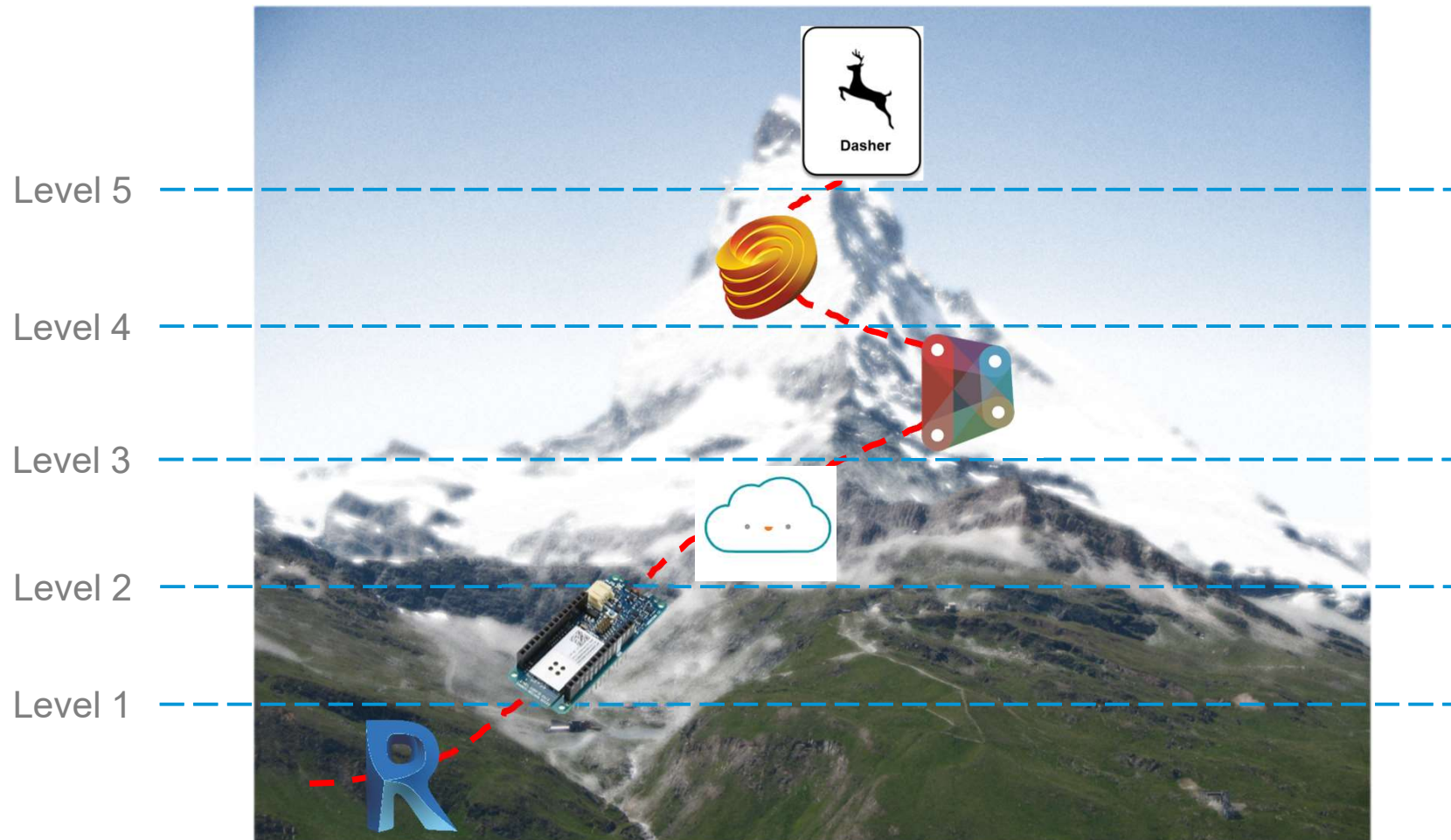
Autodesk 210 King Pier 9 Smart Bridge

Local Interior Environment Monitoring

- Temperature
- Humidity
- Motion
- CO2
- Power
- Motion
- Human activity
- Environmental
- Strain
- Computer Vision

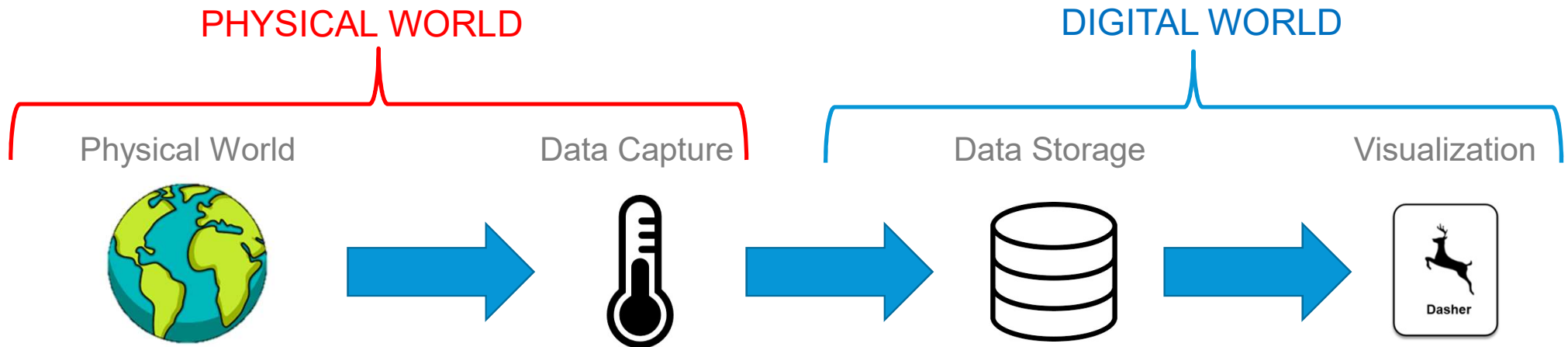


Our Journey – Storey Telling – Class Scope

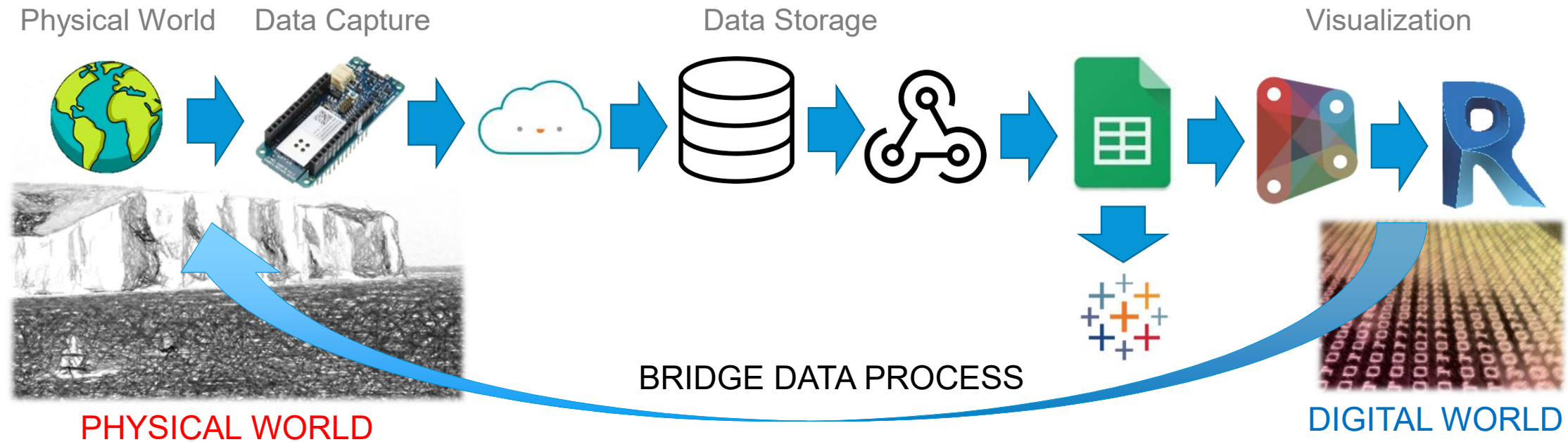


Process - Starting Point

How we "Thought" the project would work



Overall Developed Process



Overall Process Step by Step

1. Physical World: Building a Sensor

2. Data Bridge Process

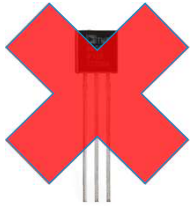
- Sensor to Cloud
- Cloud to Dashboard
- Cloud to Google Sheets
- Google Sheets to Dynamo
- Dynamo to Revit
- Revit to Dashboard

3. Digital World

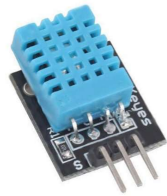
- Structured model

1. Physical World: Building a Sensor

Components used to build our sensor



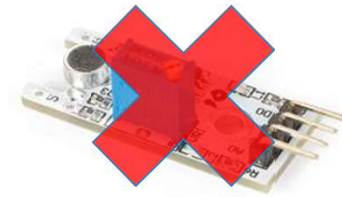
Temperature
TMP36



Temperature DHT11



Light
Photo resistor



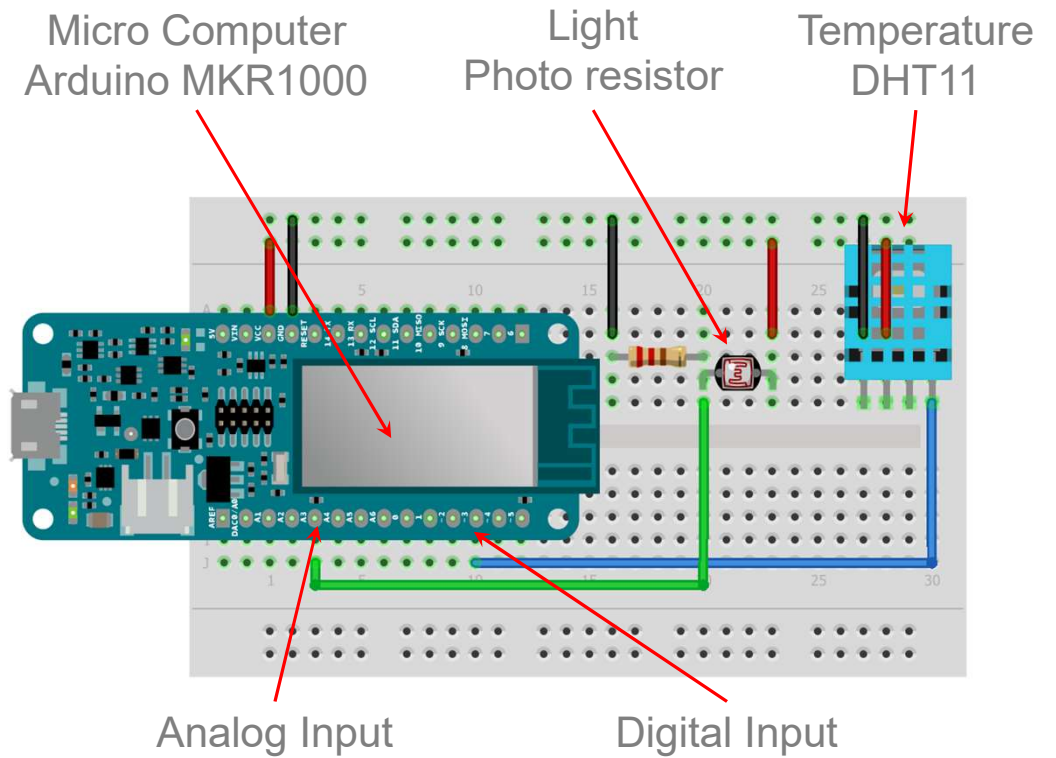
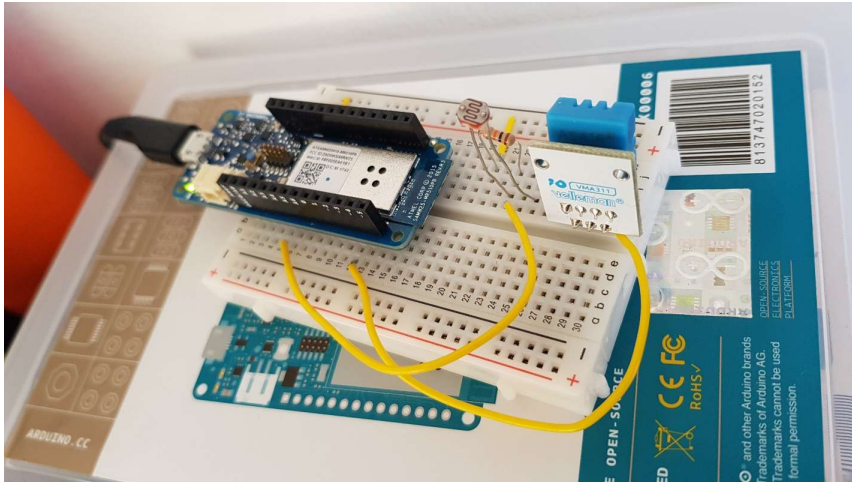
Sound
VMA 311



Micro Computer Arduino MKR1000

1. Physical World: Building a Sensor

Assembly of our sensor



2. Data Bridge Process: From Sensor To Arduino IoT Cloud

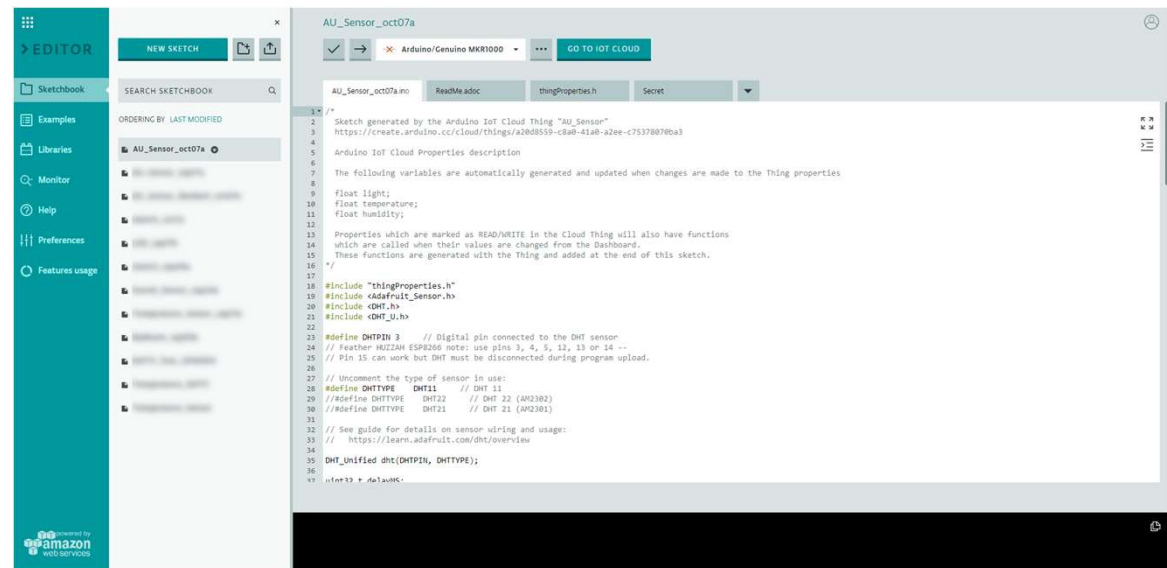
Coding the Sensor

Arduino Web Editor

- Coding the Sensor

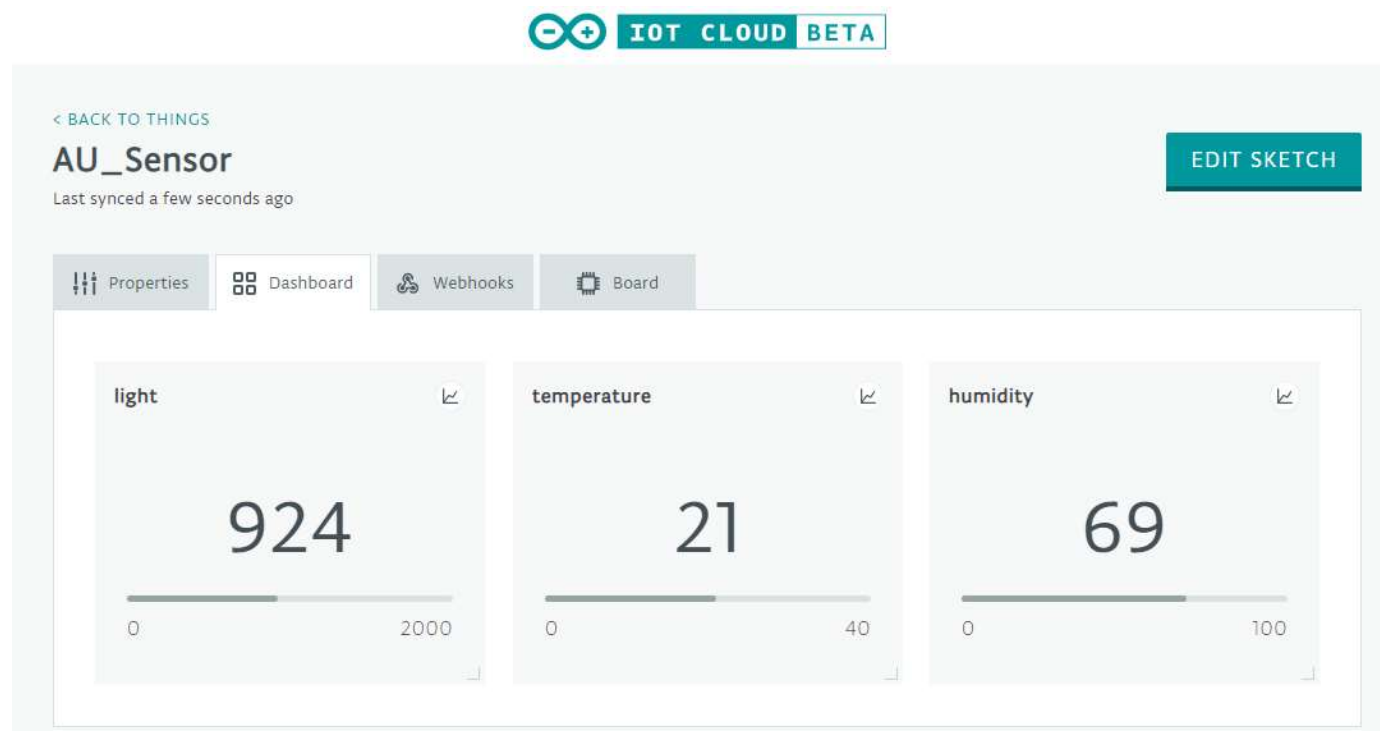
Arduino IoT Cloud

- Adding Properties to the Code
- Setting up webhook
- Dashboarding

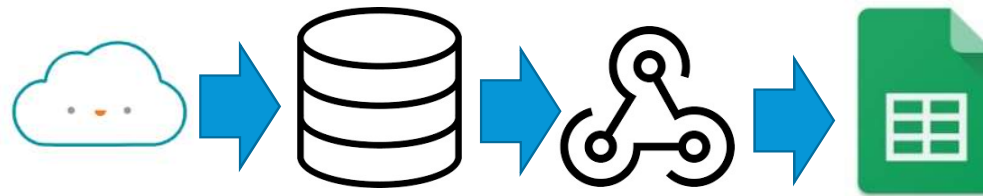


2. Data Process: Arduino IoT Cloud – Dashboard

Live data sent to cloud



2. Data Bridge Process: From Arduino IoT Cloud to Google Sheet



 IOT CLOUD BETA

[< BACK TO THINGS](#)

AU_Sensor

Last synced a minute ago

EDIT SKETCH

Properties

Dashboard

Webhooks

Board

Webhooks allow you to send and receive automated messages to other services. For example you could use a webhook to receive a notification when a property of your Thing changes. If you're new to webhooks, check this [sample project](#).

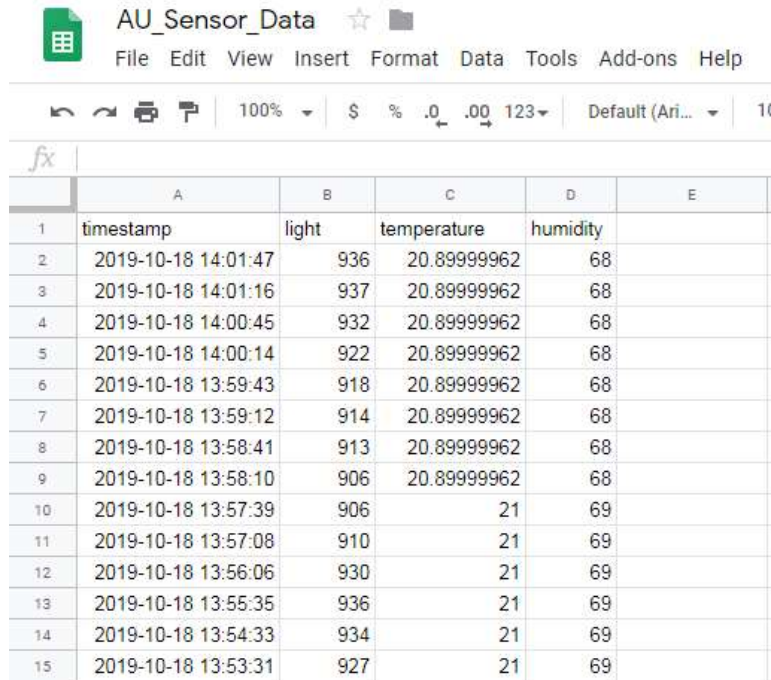
Active webhook:

`https://script.google.com/macros/s/`

Thing ID:

2. Data Bridge Process: Google Sheet Script

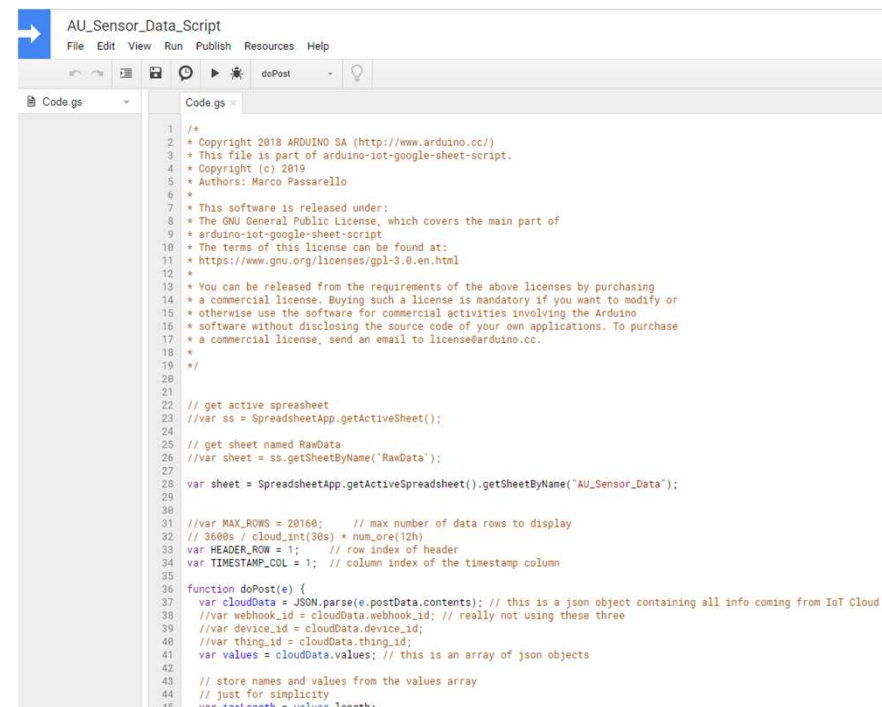
Live data sent to Google Sheets



The screenshot shows a Google Sheet titled "AU_Sensor_Data" with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Add-ons, Help) and a toolbar. The sheet contains a table with 5 columns: timestamp, light, temperature, humidity, and an empty column E. The data is as follows:

	A	B	C	D	E
1	timestamp	light	temperature	humidity	
2	2019-10-18 14:01:47	936	20.89999962	68	
3	2019-10-18 14:01:16	937	20.89999962	68	
4	2019-10-18 14:00:45	932	20.89999962	68	
5	2019-10-18 14:00:14	922	20.89999962	68	
6	2019-10-18 13:59:43	918	20.89999962	68	
7	2019-10-18 13:59:12	914	20.89999962	68	
8	2019-10-18 13:58:41	913	20.89999962	68	
9	2019-10-18 13:58:10	906	20.89999962	68	
10	2019-10-18 13:57:39	906	21	69	
11	2019-10-18 13:57:08	910	21	69	
12	2019-10-18 13:56:06	930	21	69	
13	2019-10-18 13:55:35	936	21	69	
14	2019-10-18 13:54:33	934	21	69	
15	2019-10-18 13:53:31	927	21	69	

Google Sheet



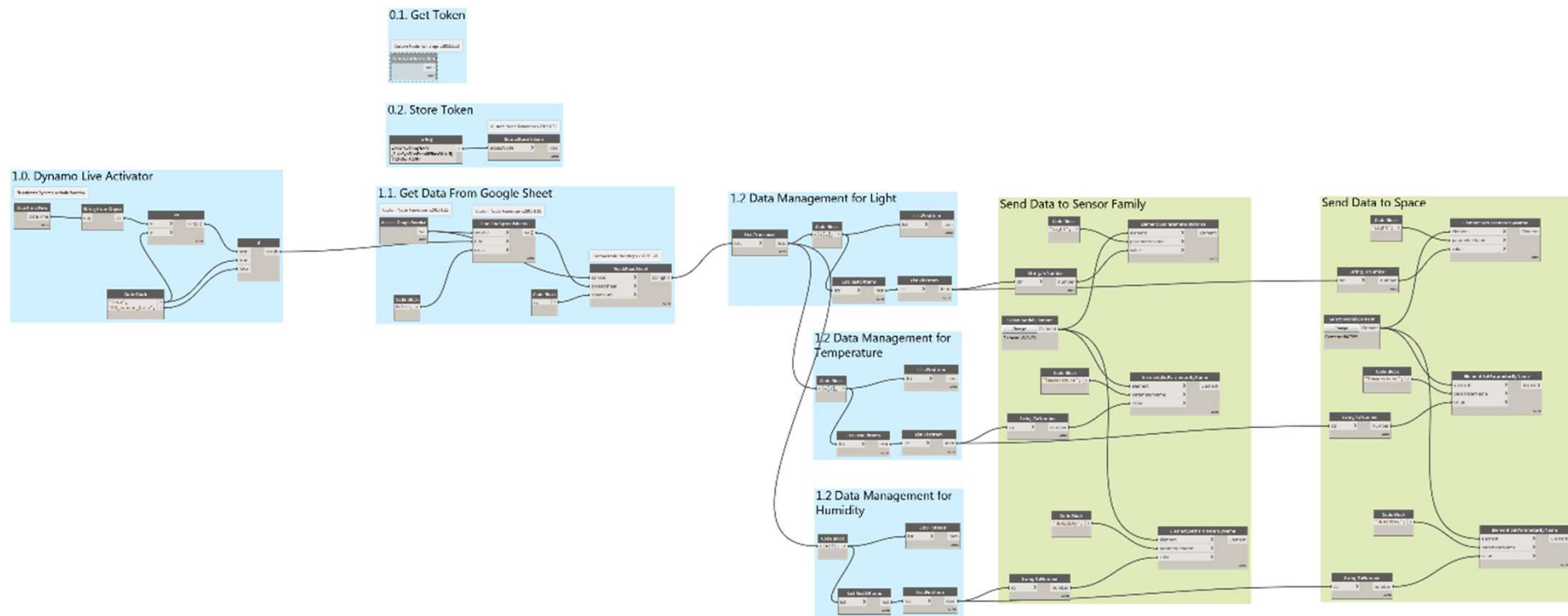
The screenshot shows a Google Script titled "AU_Sensor_Data_Script" with a menu bar (File, Edit, View, Run, Publish, Resources, Help) and a toolbar. The script is a JavaScript file that interacts with a Google Sheet. The code is as follows:

```
1 /*
2  * Copyright 2018 ARDUINO SA (http://www.arduino.cc/)
3  * This file is part of arduino-iot-google-sheet-script.
4  * Copyright (c) 2019
5  * Authors: Marco Passarello
6  *
7  * This software is released under:
8  * The GNU General Public License, which covers the main part of
9  * arduino-iot-google-sheet-script
10 * The terms of this license can be found at:
11 * https://www.gnu.org/licenses/gpl-3.0.en.html
12 *
13 * You can be released from the requirements of the above licenses by purchasing
14 * a commercial license. Buying such a license is mandatory if you want to modify or
15 * otherwise use the software for commercial activities involving the Arduino
16 * software without disclosing the source code of your own applications. To purchase
17 * a commercial license, send an email to license@arduino.cc.
18 *
19 */
20
21
22 // get active spreadsheet
23 //var ss = SpreadsheetApp.getActiveSheet();
24
25 // get sheet named RawData
26 //var sheet = ss.getSheetByName('RawData');
27
28 var sheet = SpreadsheetApp.getActiveSpreadsheet().getSheetByName("AU_Sensor_Data");
29
30
31 //var MAX_ROWS = 2048; // max number of data rows to display
32 // 3600s / cloud_int(30s) * num_ore(12h)
33 var HEADER_ROW = 1; // row index of header
34 var TIMESTAMP_COL = 1; // column index of the timestamp column
35
36 function doPost(e) {
37   var cloudData = JSON.parse(e.postData.contents); // this is a json object containing all info coming from IoT Cloud
38   //var webhook_id = cloudData.webhook_id; // really not using these three
39   //var device_id = cloudData.device_id;
40   //var thing_id = cloudData.thing_id;
41   var values = cloudData.values; // this is an array of json objects
42
43   // store names and values from the values array
44   // just for simplicity
45   var length = values.length;
```

Google Script

2. Data Bridge Process: From Google Sheet to Dynamo-Revit

The bridge between data and the model.

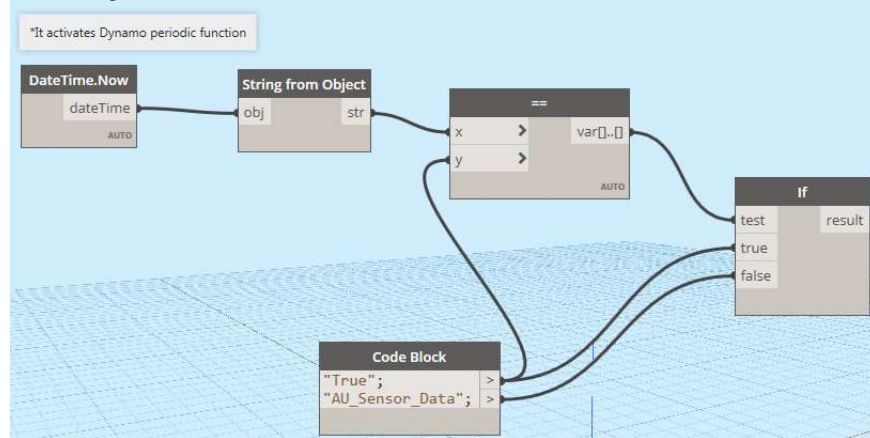


2. Data Bridge Process: Key Parts of Dynamo Graph

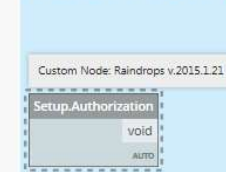
Live activator to use the periodic function

Raindrops package to get data from Google Sheets

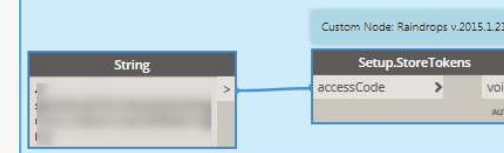
1.0. Dynamo Live Activator



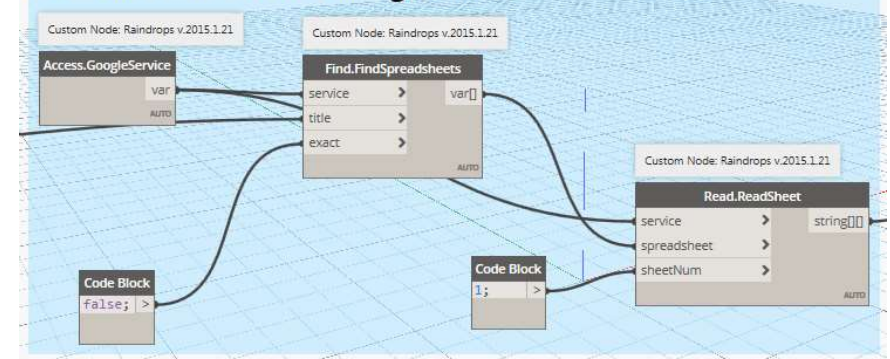
0.1. Get Token



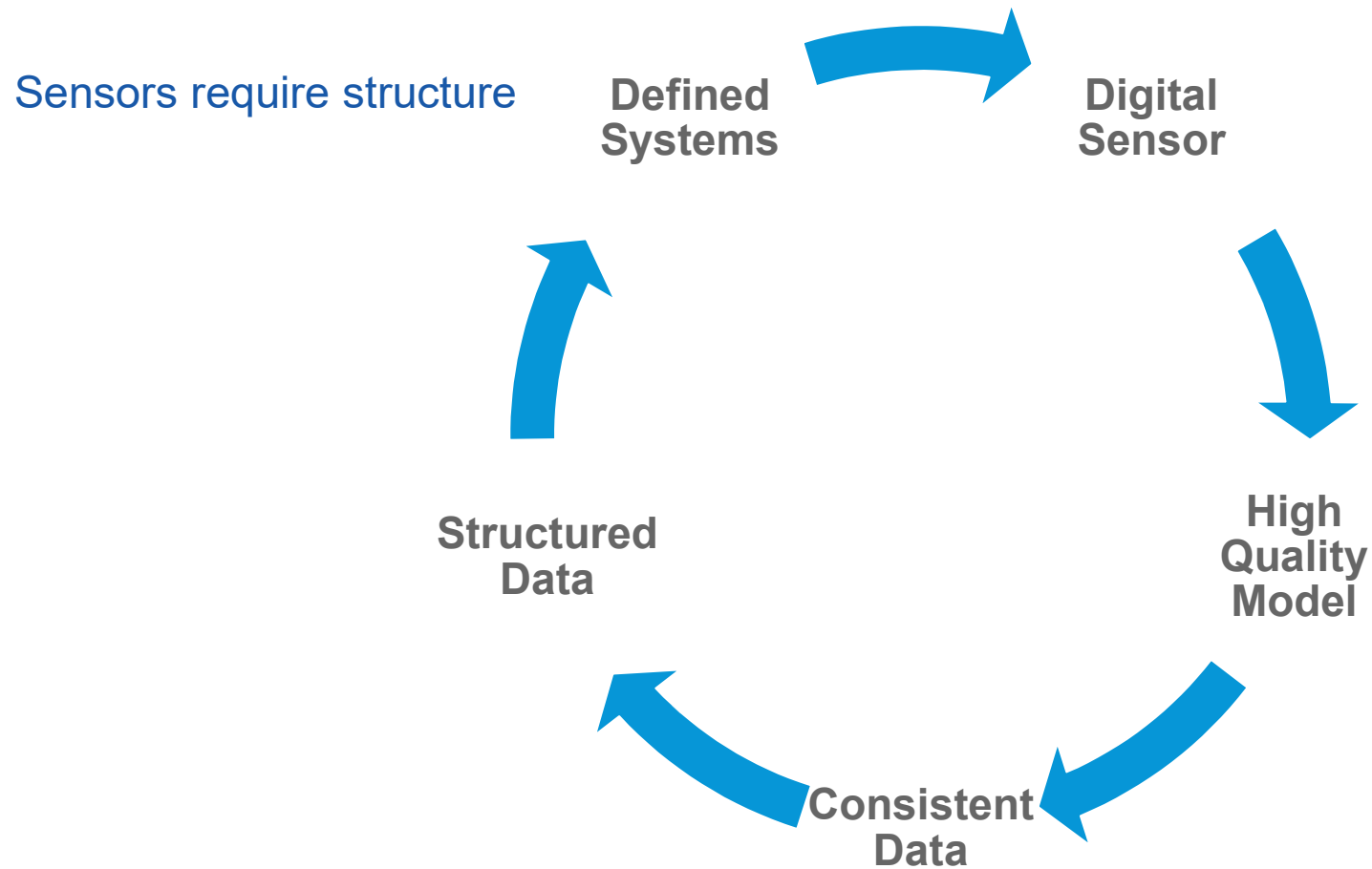
0.2. Store Token



1.1. Get Data From Google Sheet

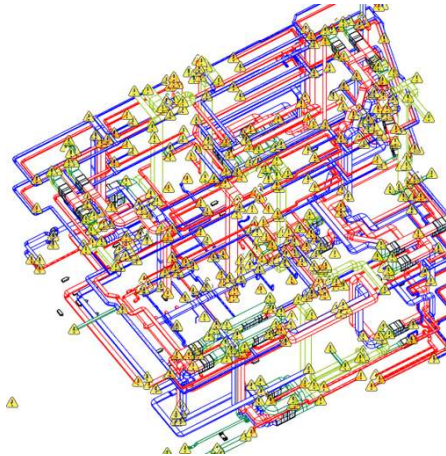
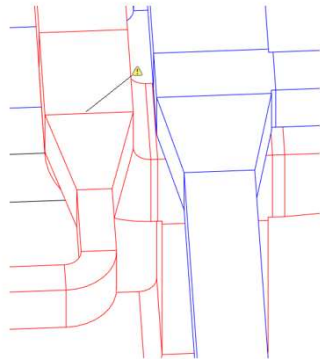
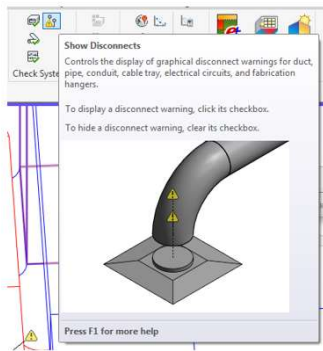


3. Digital World: Well Structured – MEP Model



3. Digital World: Well Structured – MEP Model

Good Modelling Practices



Systems & MEP Spaces

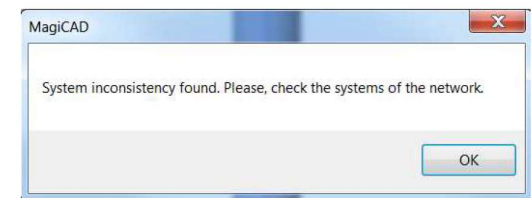
System Browser - B7H_K08_F2_Ventilation_jnc@ram...

View: Systems All Disciplines

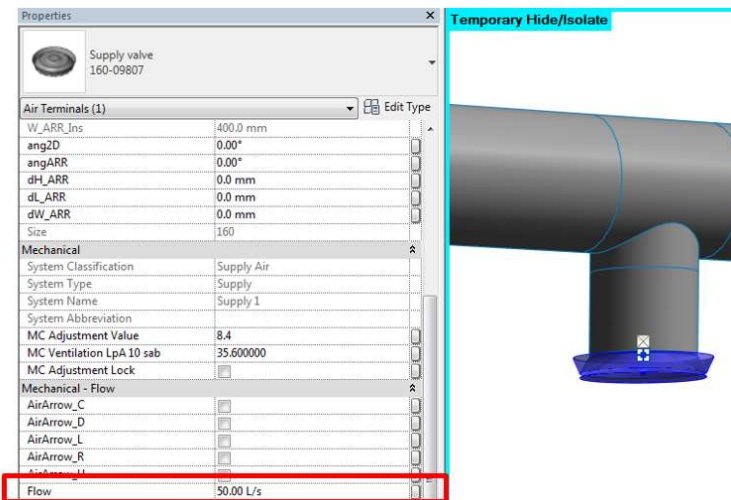
Systems	Flow	Size	Spa
Unassigned (0 items)			
Mechanical (229 systems)			
Ventilation, Afkast			
Ventilation, Indblesning			
1 VE01 og VE02	Not...		
1 VE03	Not...		
1 VE04	Not...		
1 VE07	Not...		
1 VE08	Not...		
1 VE09	Not...		
1 VE10	Not...		
1 VE11	Not...		
1 VE12	Not...		
1 VE13	Not...		
1 VE14	Not...		
1 VE15	Not...		
1 VE16	Not...		
1 VE17	Not...		
1 VE18	Not...		
1 VE19	Not...		
1 VE20	Not...		
1 VE21	Not...		
1 VE26	Not...		
VI 1	Not...		

Mechanical

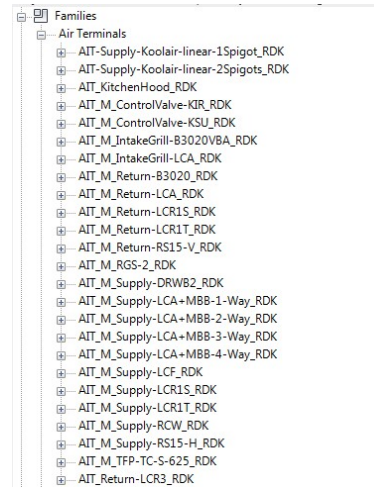
System Classification	Supply Air
System Type	
System Name	
System Abbreviation	VENT I



Data Structure



Data Consistency



3. Digital World: Revit Sensor Family

Building a Revit sensor family

Data Device

Family type name as unique identifier

Instance parameters for receiving data

Type parameters to define sensor

Simple form as a placeholder

Each virtual sensor needs to be unique.

Family Types

Type name: Light-Temp-RelHumid_RHO1182_Lux-C-%_RHO1182-01

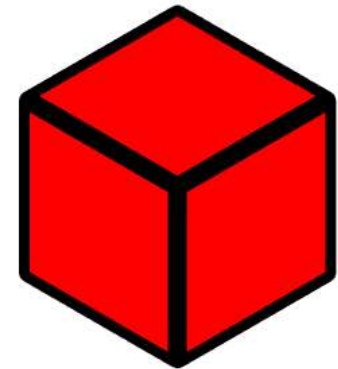
Search parameters

Parameter	Value	Formula	Lock
Constraints			
Default Elevation	0.0	=	
Data			
Humidity (default)	0.000000	=	
Light (default)	0.000000	=	
Light Data (default)	https://docs.google.co	=	
Sensor_ID	RHO1182-01	=	
Sensor_Location	RHO1182	=	
Sensor_Type	Light-Temp-RelHumid	=	
Sensor_Units	Lux-C-%	=	
Temperature (default)	0.000000	=	
Temperature and Humidit	https://docs.google.co	=	
Identity Data			

Manage Lookup Tables

How do I manage family types?

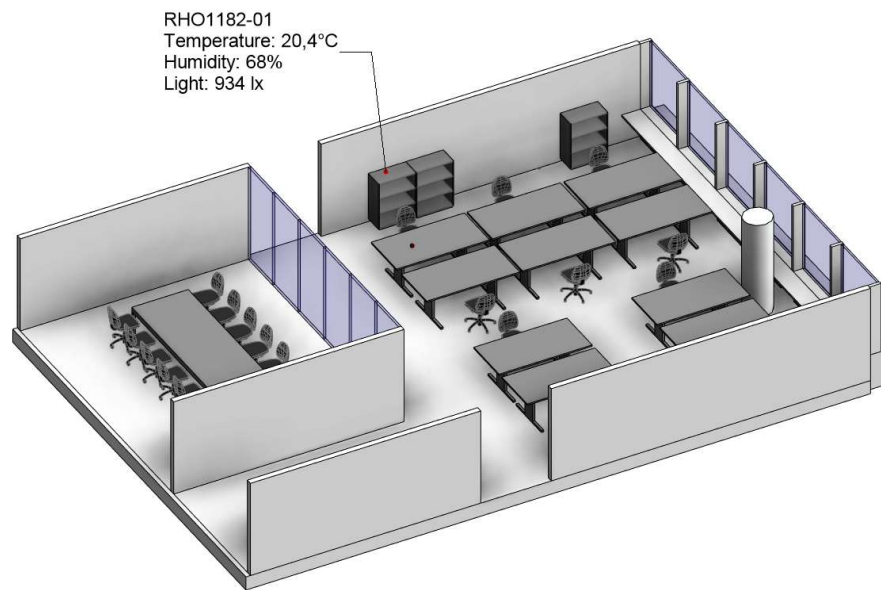
OK Cancel Apply



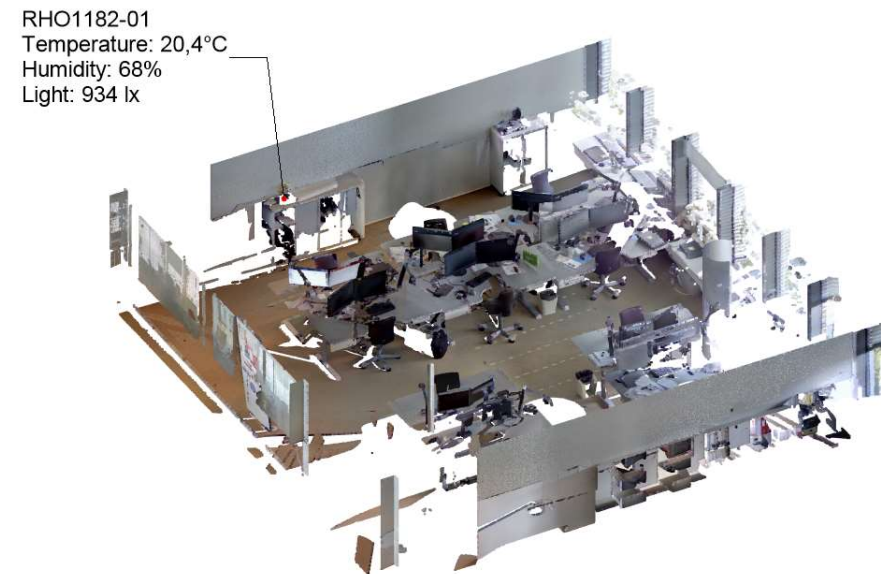
DEMO

"BRINGING TOGETHER MEP MODELS AND IoT SENSORS"

Study Case : Digital Twin – Our Office Room

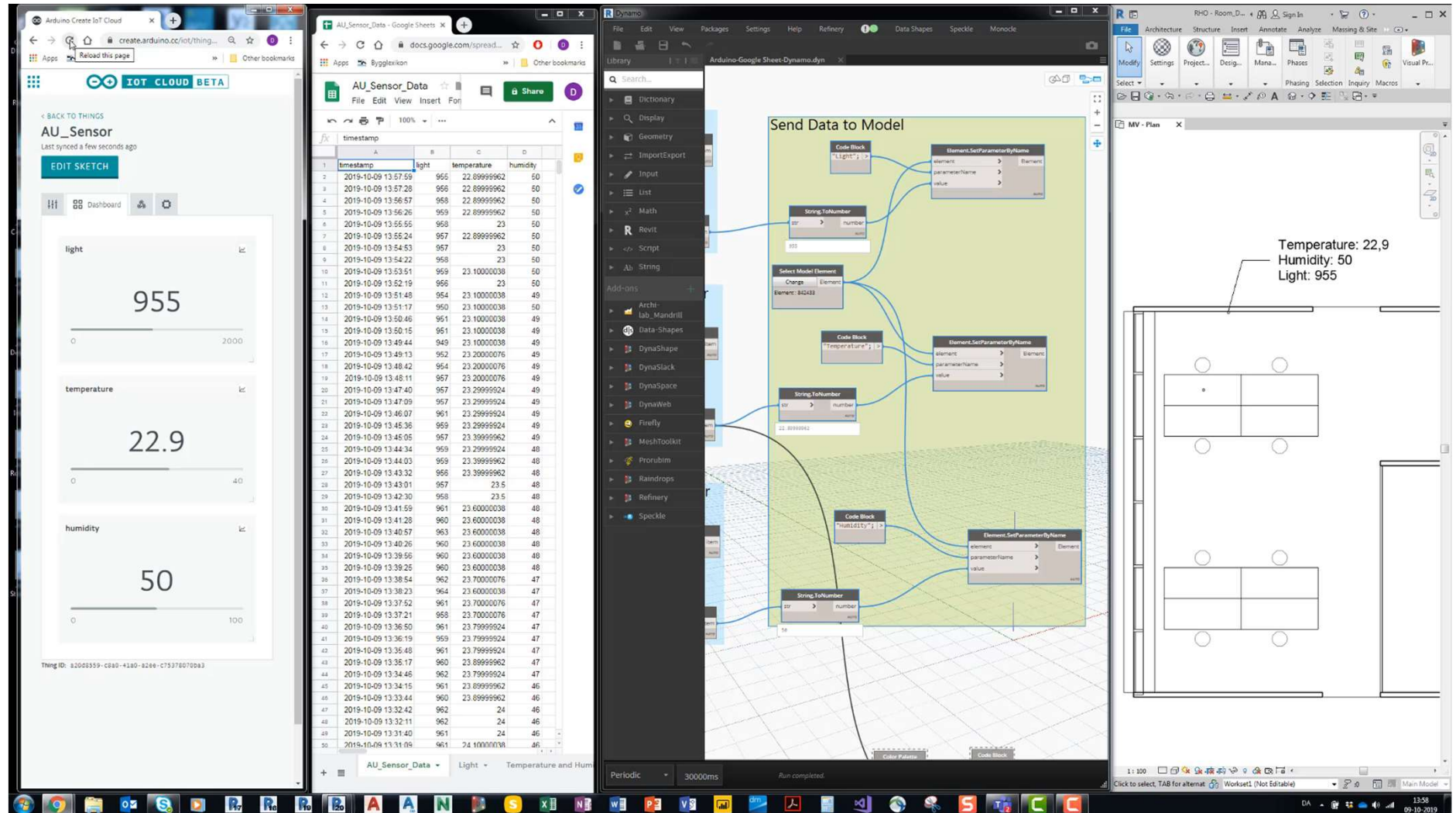


Revit Model

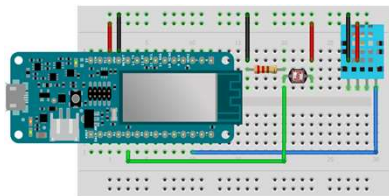


3D Scan

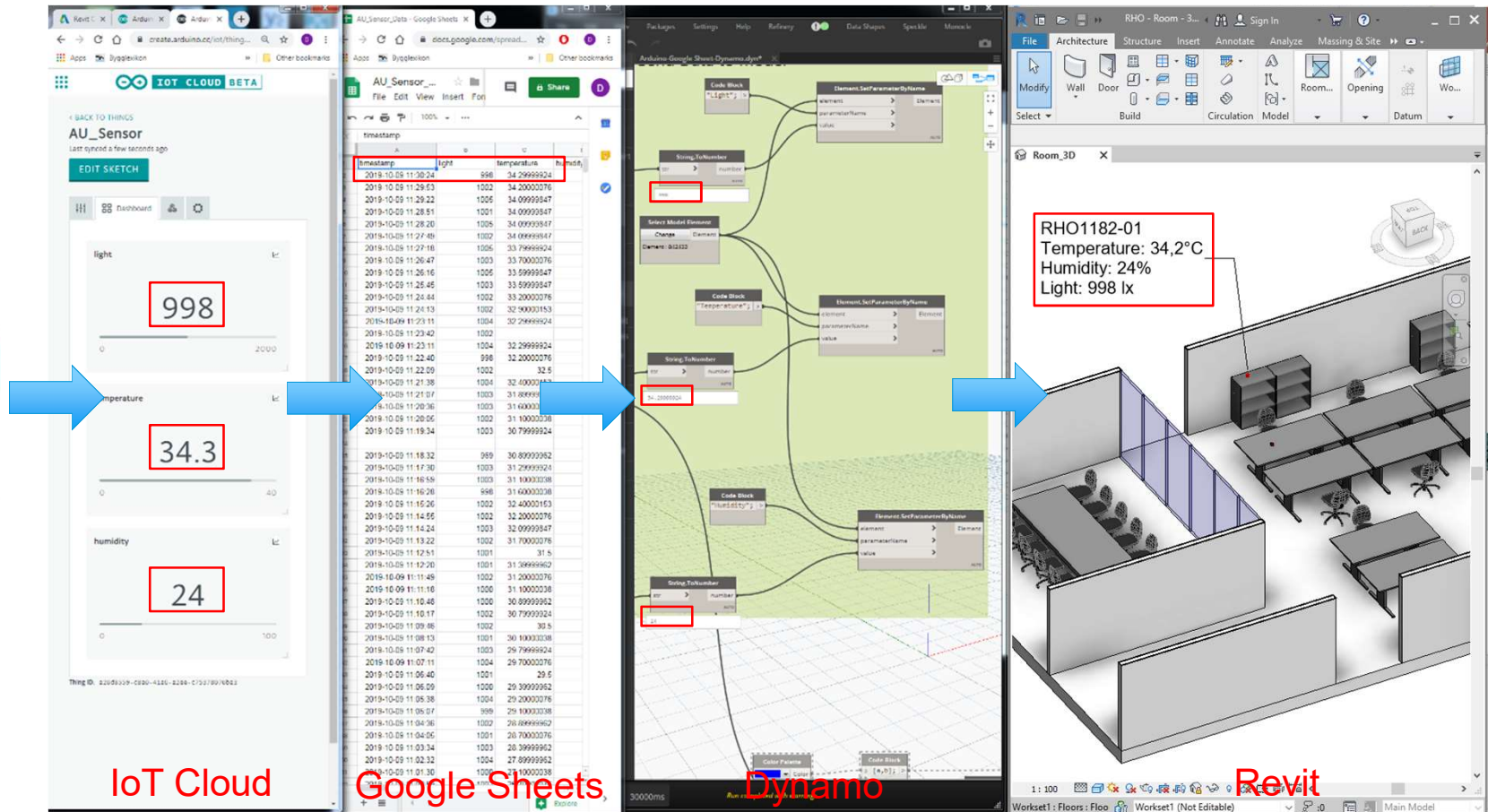
Study Case demo



Process Summary



Arduino Sensor



3. Learn how to visualize live IoT data with BI dashboards

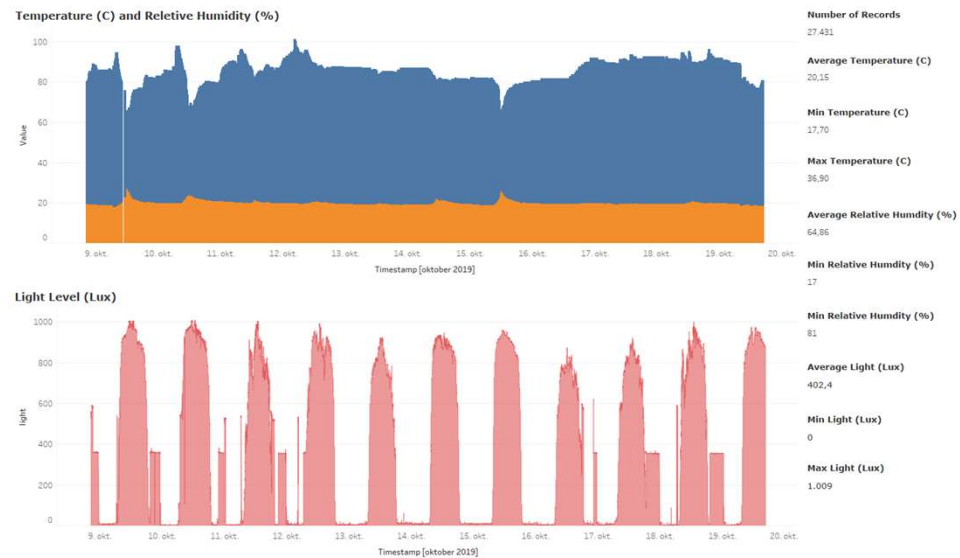


Dashboarding

The purpose of visualization is insight, not pictures

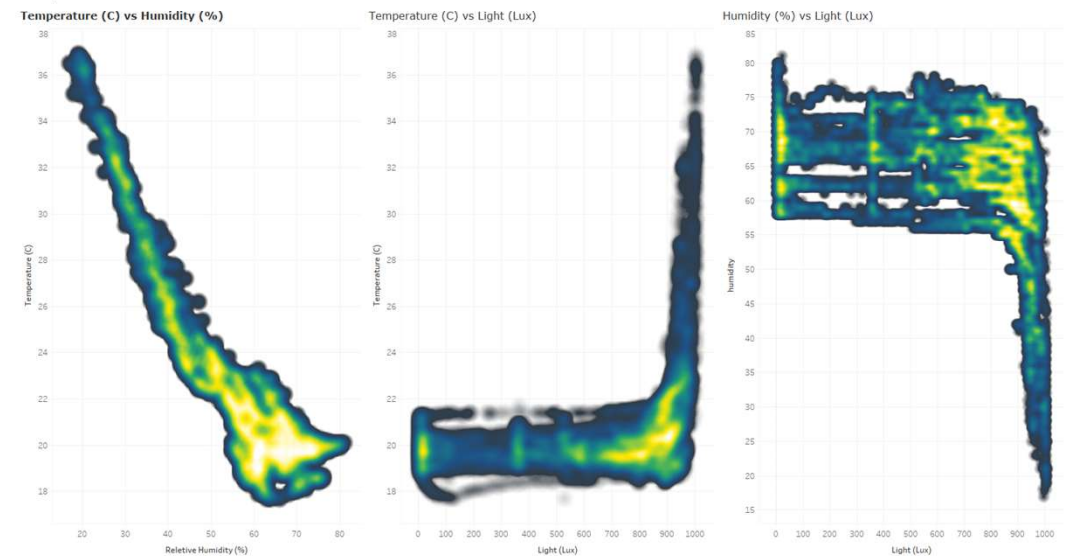
Decision making is a critical part

General Dashboard



Visualizaton of Temperature, Humidity and Light

Comparative Dashboard

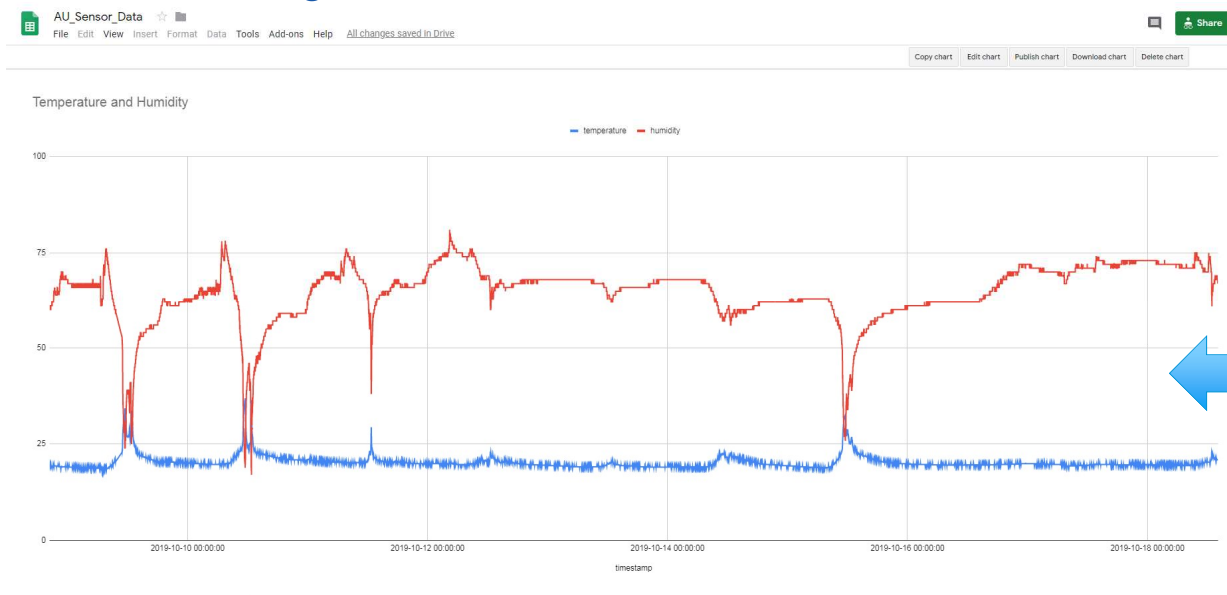


Analysis of Temperature Humidity and Light

Dashboard Loop

Linking the Revit sensor back to the physical world

- Going back to where the data lives



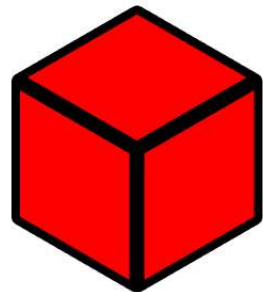
Google Sheets

The image shows the "Family Types" window in Revit. The "Type name" is "Light-Temp-RelHumid_RHO1182_Lux-C-%_RHO1182-01". The "Search parameters" section is empty. The "Parameters" table has columns for "Parameter", "Value", "Formula", and "Lock". The "Data" section contains the following parameters:

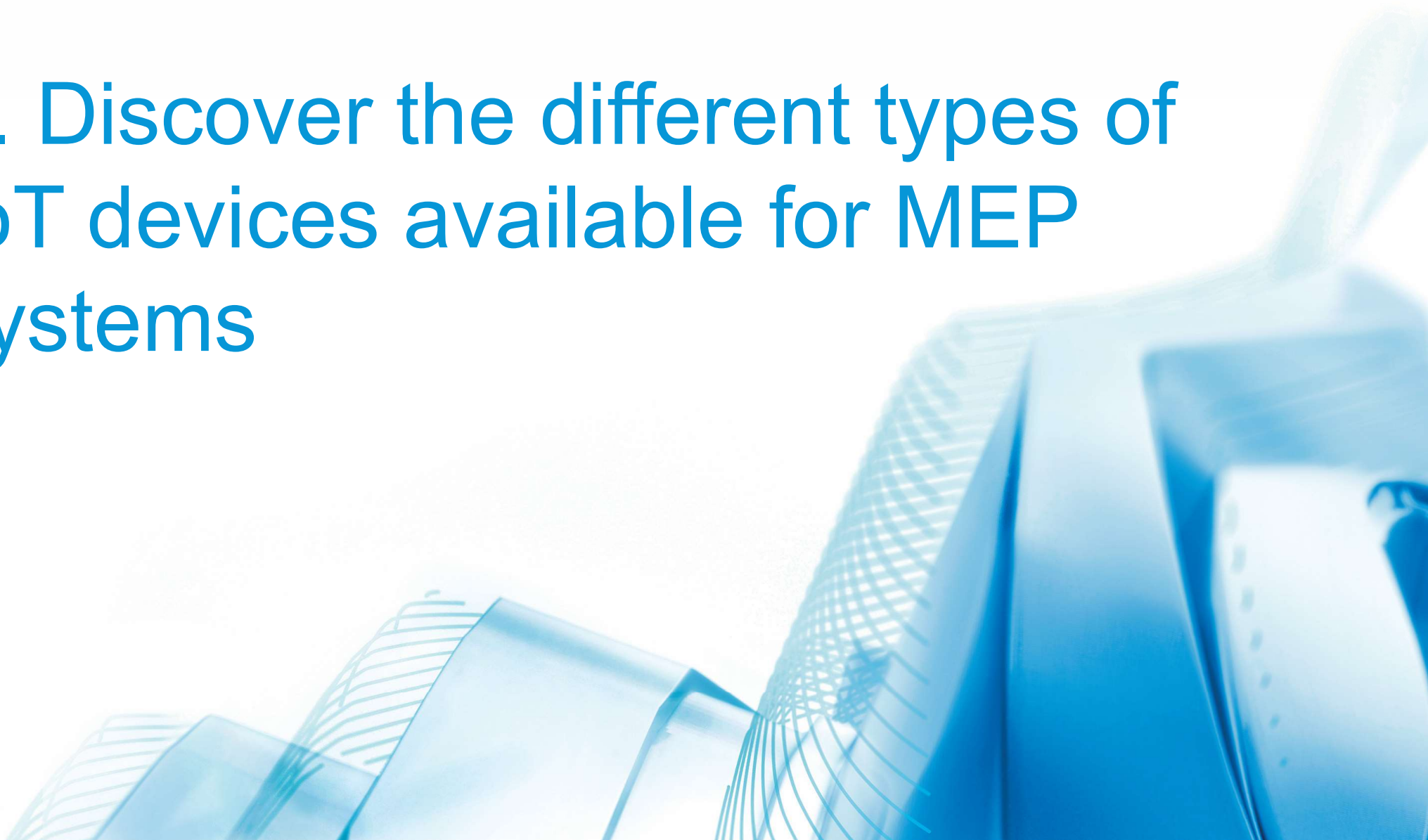
Parameter	Value	Formula	Lock
Default Elevation	0.0	=	
Humidity (default)	0.000000	=	
Light (default)	0.000000	=	
Light Data (default)	https://docs.google.co	=	
Sensor_ID	RHO1182-01	=	
Sensor_Location	RHO1182	=	
Sensor_Type	Light-Temp-RelHumid	=	
Sensor_Units	Lux-C-%	=	
Temperature (default)	0.000000	=	
Temperature and Humidity	https://docs.google.co	=	

The "Identity Data" section is empty. The "Manage Lookup Tables" button is at the bottom right. A red box highlights the "Temperature and Humidity" parameter row.

Revit Family



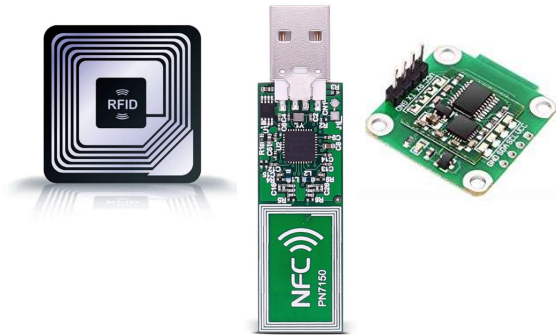
4. Discover the different types of IoT devices available for MEP systems



Types of Available Sensors

Short Distance Sensors

- RFID
- NFC
- Bluetooth
- WUSB
- Infrared



Medium Distance Sensors

- ZigBee
- EnOcean
- WiFi – 2,4 GHz
- WiFi – 5 GHz
- RF



Long Distance Sensors

- NB-IoT (Narrow Band IoT)
- SigFox
- LoRaWAN
- 4G
- 5G



Wrap Up & Conclusion



Digital Twin Challenges

- Accessing the Sensors
 - Maintenance – Verification functioning as intended
 - Calibration – Many sensors need to be periodically calibrated
 - Locating – Finding the sensors after they have been installed
- Power the sensors and data in return
 - Cabling – Extra cost to install power and data to and from the sensors.
 - Batteries – Changing and the environmental impact of disposal.
- Planning
 - Measuring – What needs to be measured? How to measure?
 - System – How to get access to the data? Structure of the data?

Digital Twin Challenges

- Data
 - Accessing the data
 - Data structure and compatibility
 - Open and closed systems
 - Security
 - Data encryption
 - Access privileges and the clear definition of user roles

Key findings

- Well Data Structured Model is Required
- Facilities Management Data Requirements about MEP sensors should be planned before starting with the project
- Data Management Strategy is required for scaling – You can't log data every second
- Dilemma between building owner and facilities management operator interest (who paid for digital twin model)

Digital Twin – Benefits

End point control – Occupant centric control of their environment

Improved system level efficiency

Reduction in energy usage = Reduction in CO₂ and greenhouse gas emissions

- Almost 50% of pollution comes from powering buildings.

Through real-time feedback occupants become aware of their impact on the environment.

Key learnings

- Everyone can do a digital twin model.
- The challenge is not the technology, but the management of data.
- Digital Twin should be planned from early phases.
- Digital Twin model should have a purpose and deliver value to the client.
- Smart Buildings is big opportunity for MEP engineers to offer new design services.

Thanks for your attention!

Questions?

- David A. Fink – Henning Larsen Architects
dfi@henninglarsen.com



 @finkBIM

- Alejandro Mata – Ramboll
- almat@ramboll.dk



 @alejandromataio

RAMBOLL

**Henning
Larsen —**



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